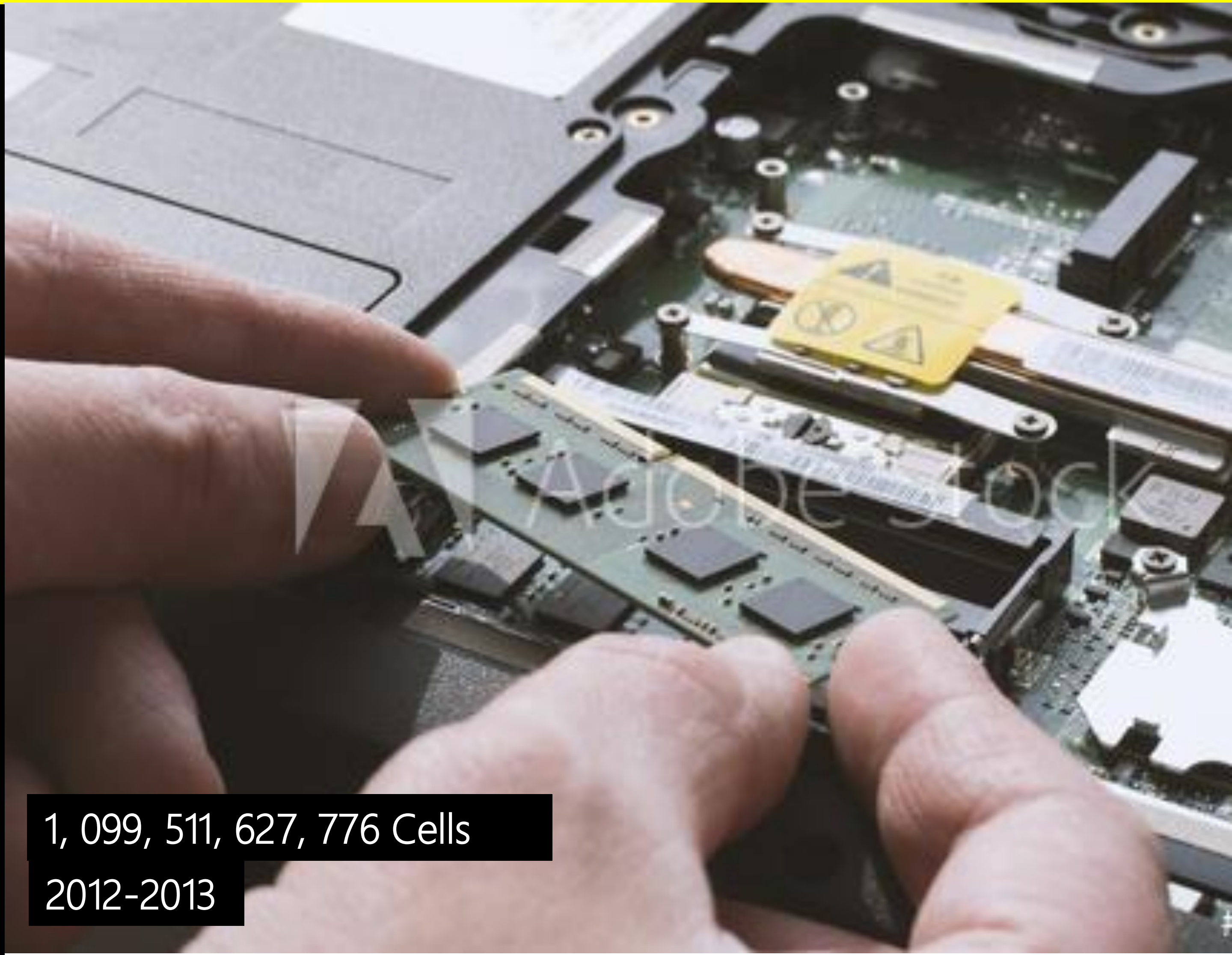
$16\ G = 16\times2^{10}\ (M) = 16\times2^{10}\times2^{10}\ (K) = 16\times2^{10}\times2^{10}\times2^{10}$ Memory Locations

How many Cells?

$(2^4=16)\times2^{10}\times2^{10}\times2^{10}\times(2^6=64\text{-bit Data Bus}) = 2^{40}$

1 Cell

1958–1959

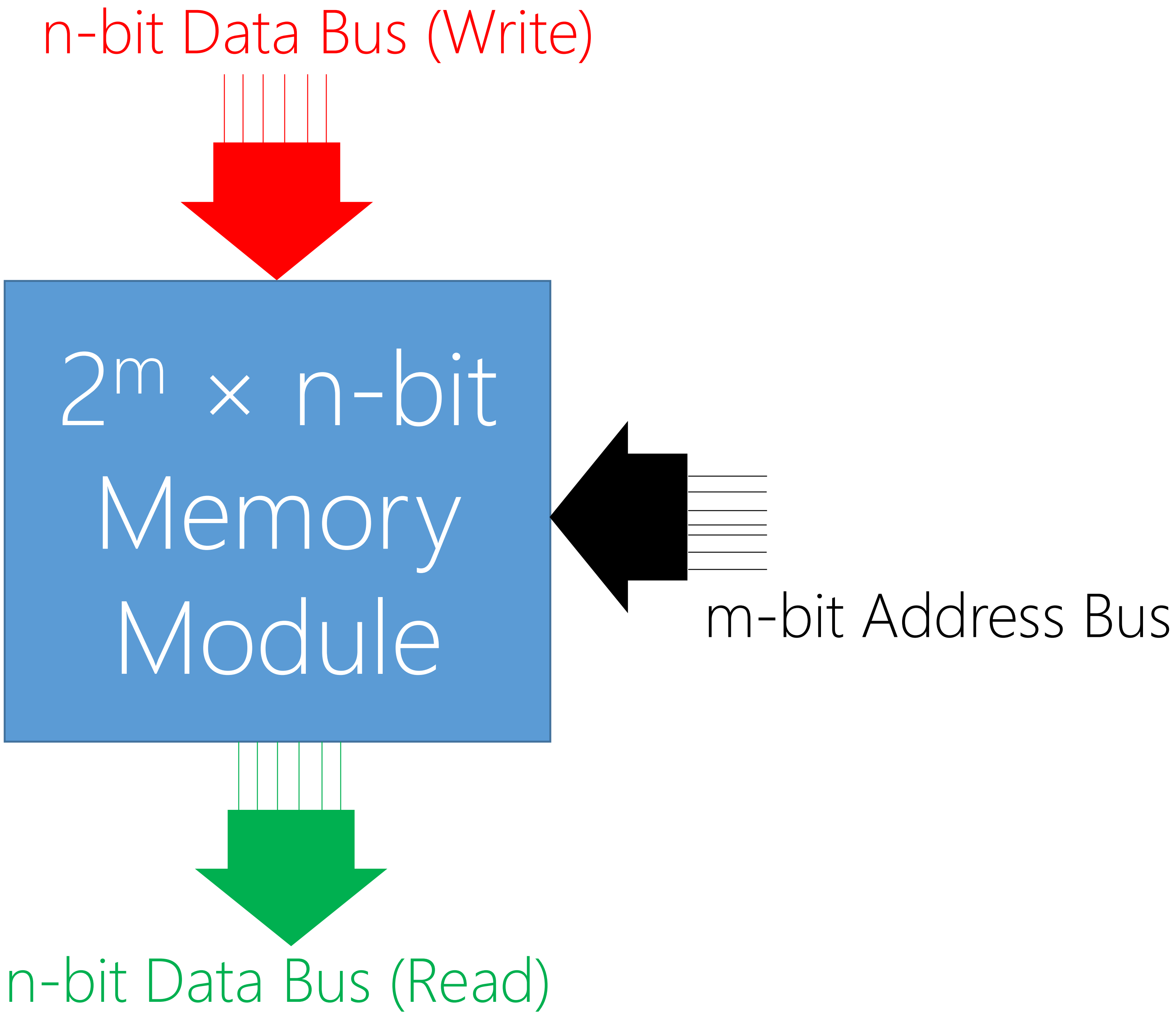


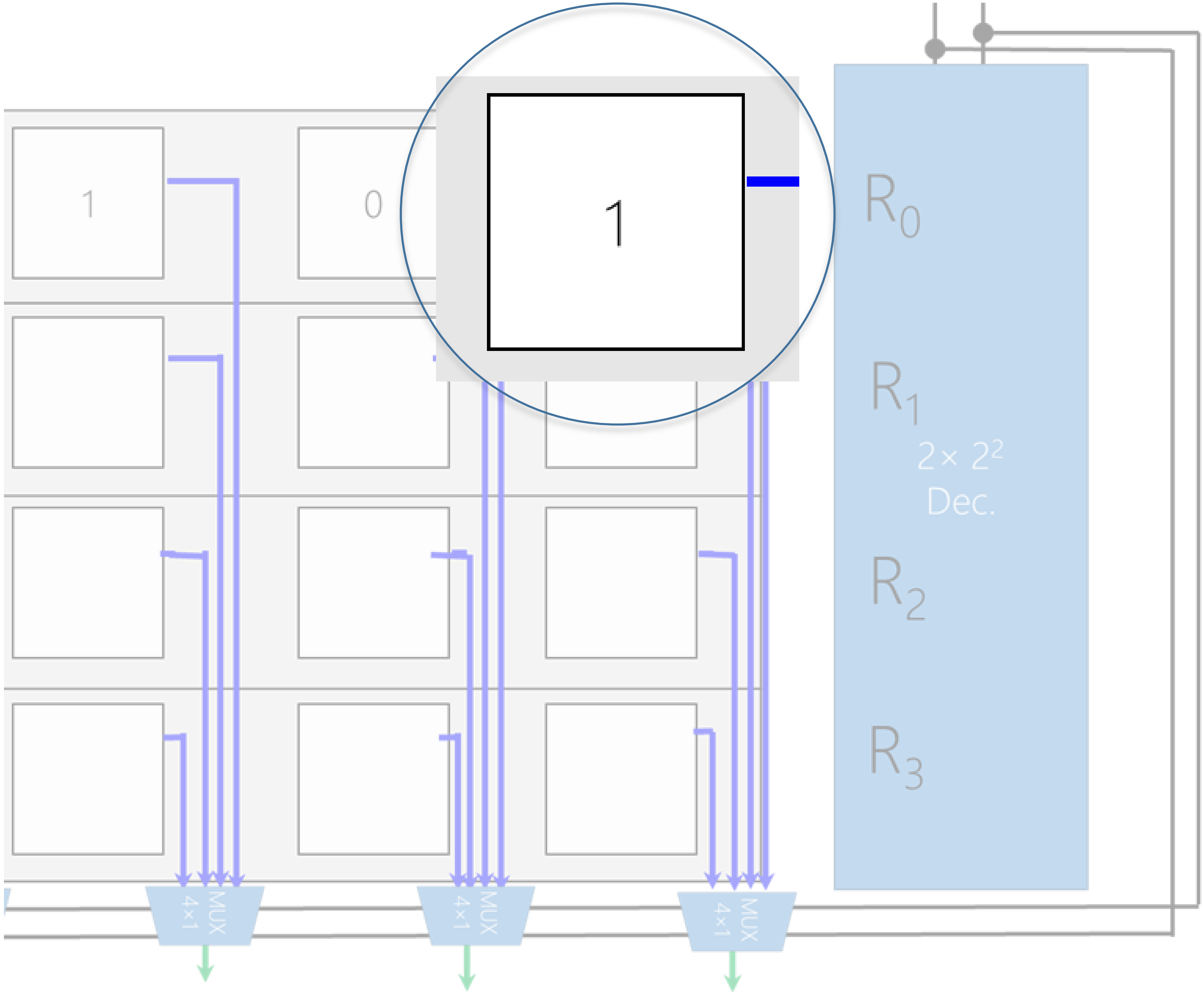
1, 099, 511, 627, 776 Cells

2012-2013

F2024: A Digital Odyssey

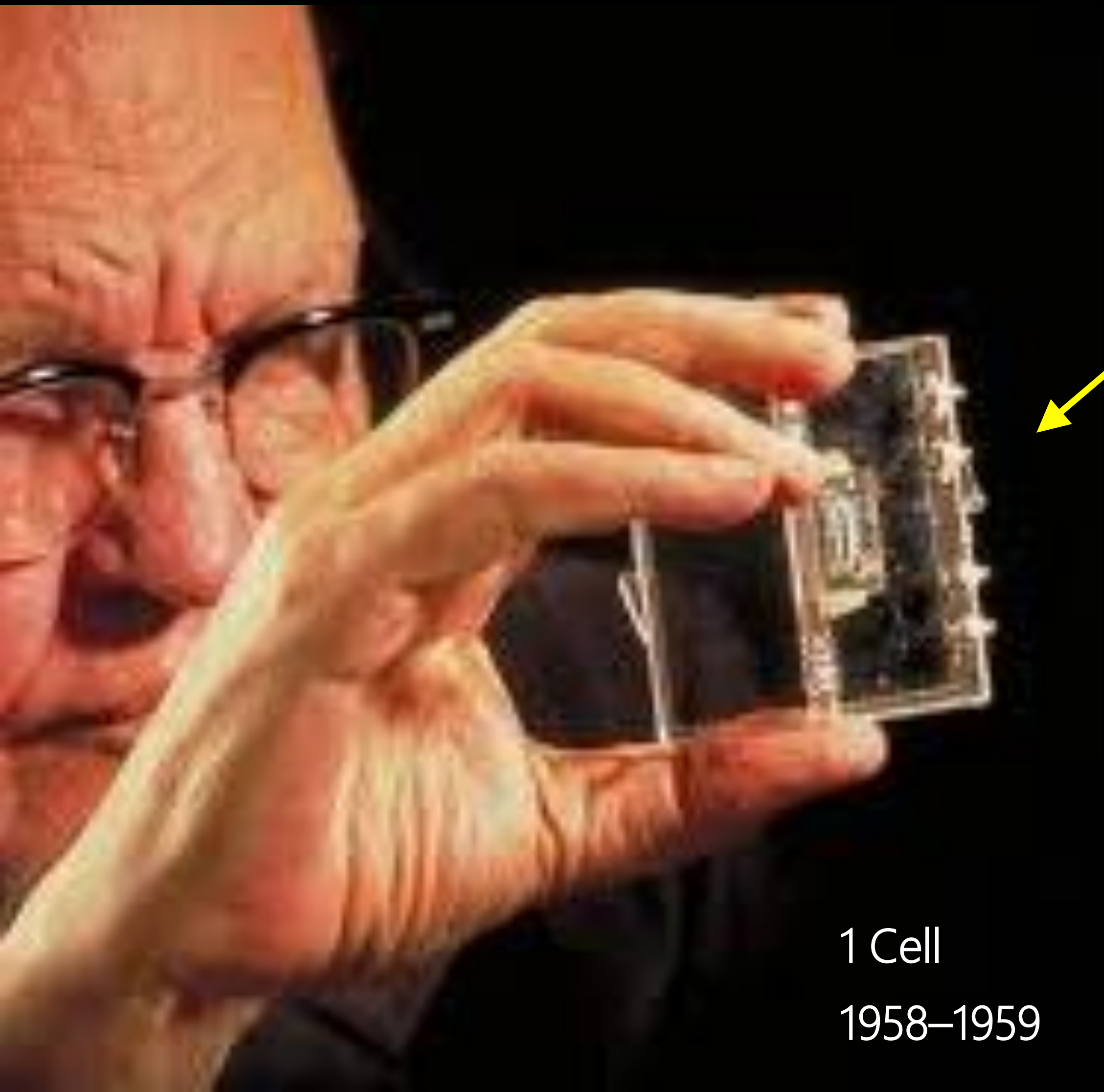
The Ultimate Trip.


n-bit Data Bus (Write)
$2^m \times$ n-bit
Memory
Module
m-bit Address Bus
n-bit Data Bus (Read)



1
0
1
R0
R1
2× 2^2
Dec.
R2
R3
MUX 4×1
MUX 4×1
MUX 4×1

What is that cell?

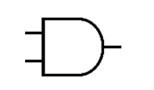


1 Cell

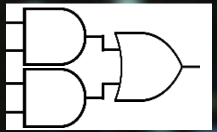
1958–1959

Number Systems | $(12)_{10} \rightarrow (1100)_2$

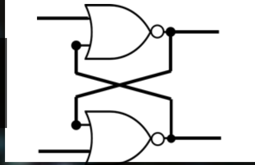
Logic Gates |



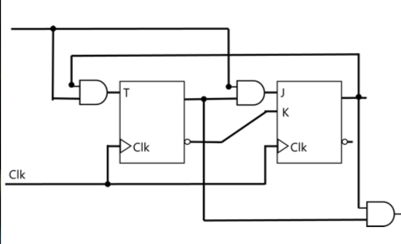
Combinational Logic |

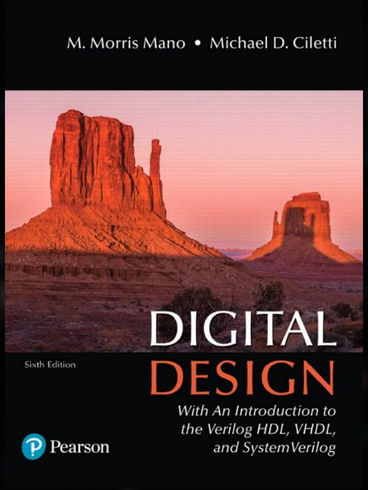


Flip-Flop |



Sequential Logic |



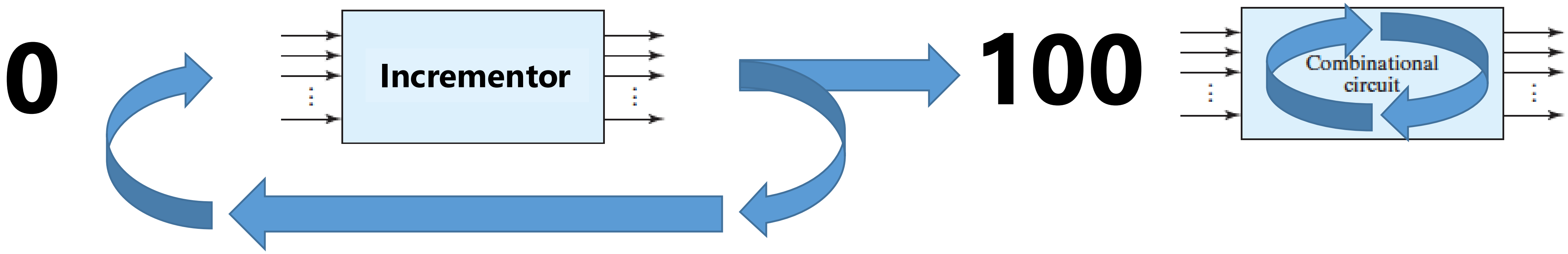


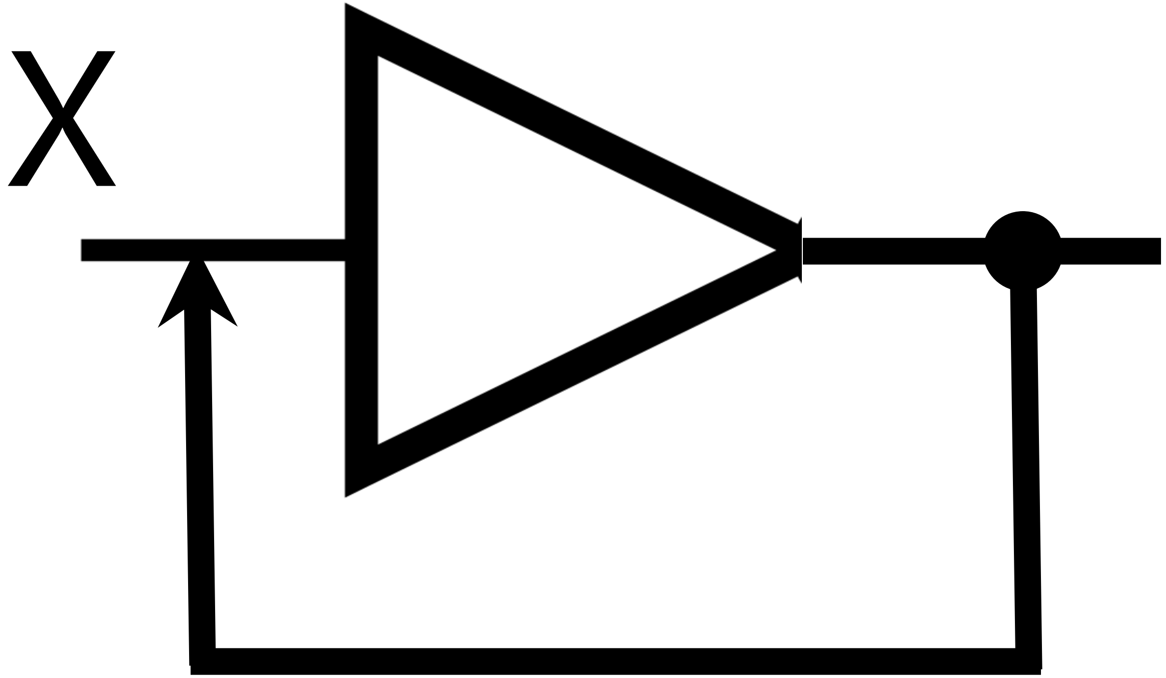
Chapter 5
Synchronous Sequential Logic

Memento, Christopher Nolan
Guy Pearce, Carrie-Anne Moss, Joe Pantoliano
September 2000
Budget $4.5 m
Box office $40 m

Sequential Logic

Counter: 0 → 1 → 2 → …

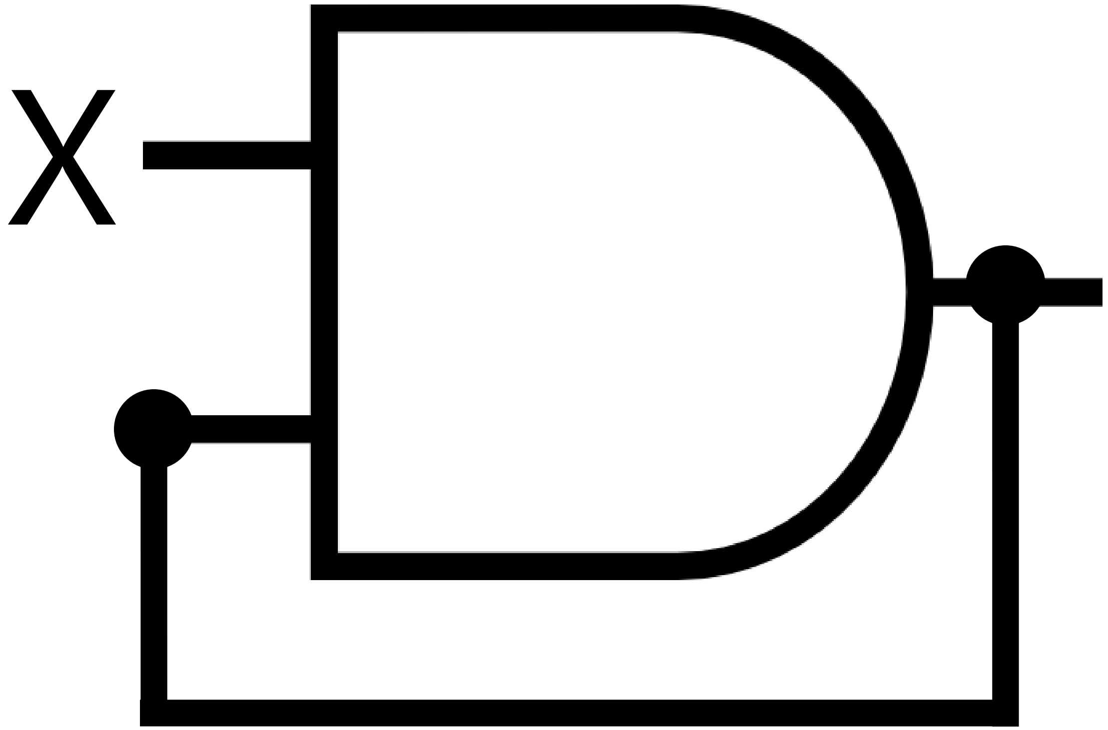




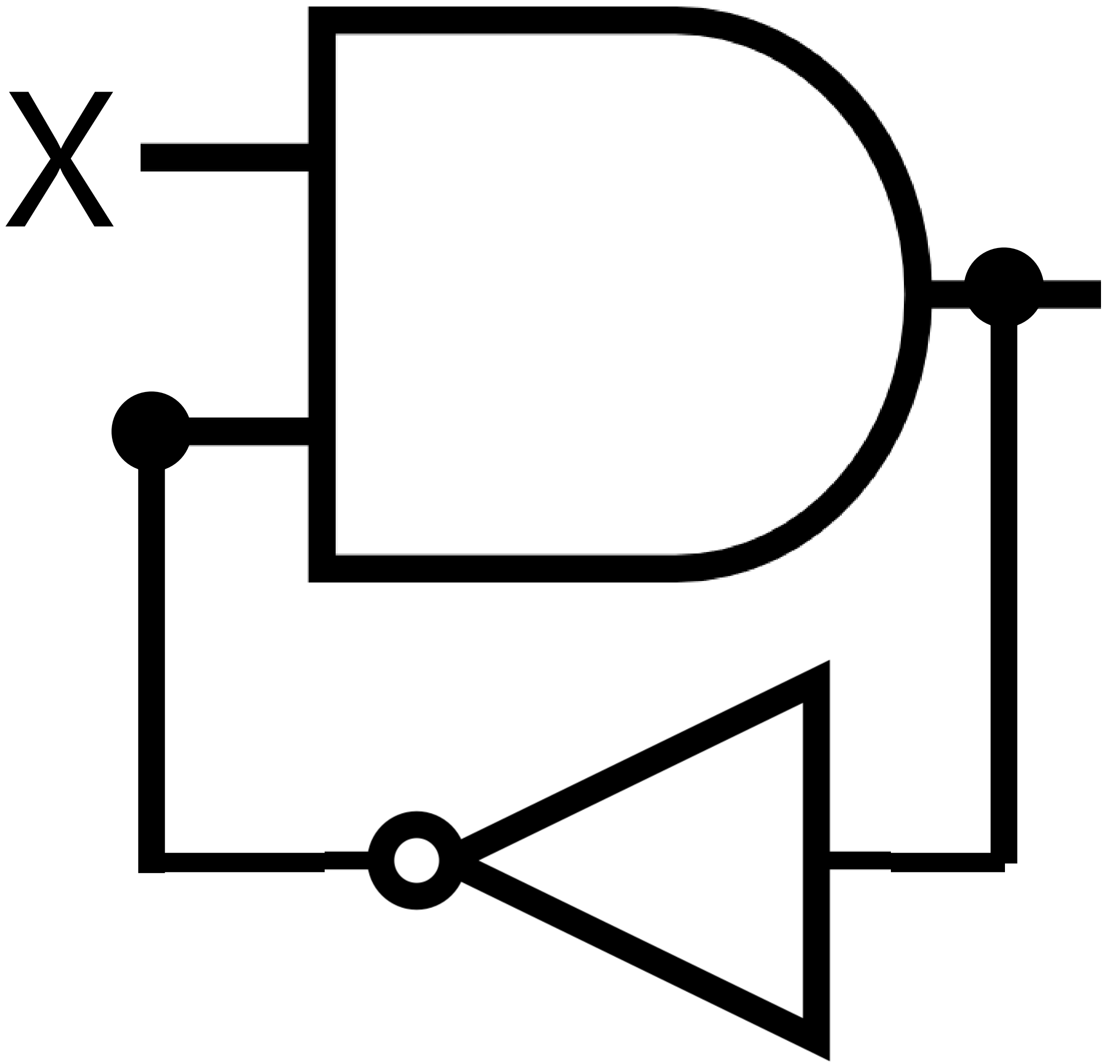
This design is logically incorrect! Why?

Buffer gate accepts one input.

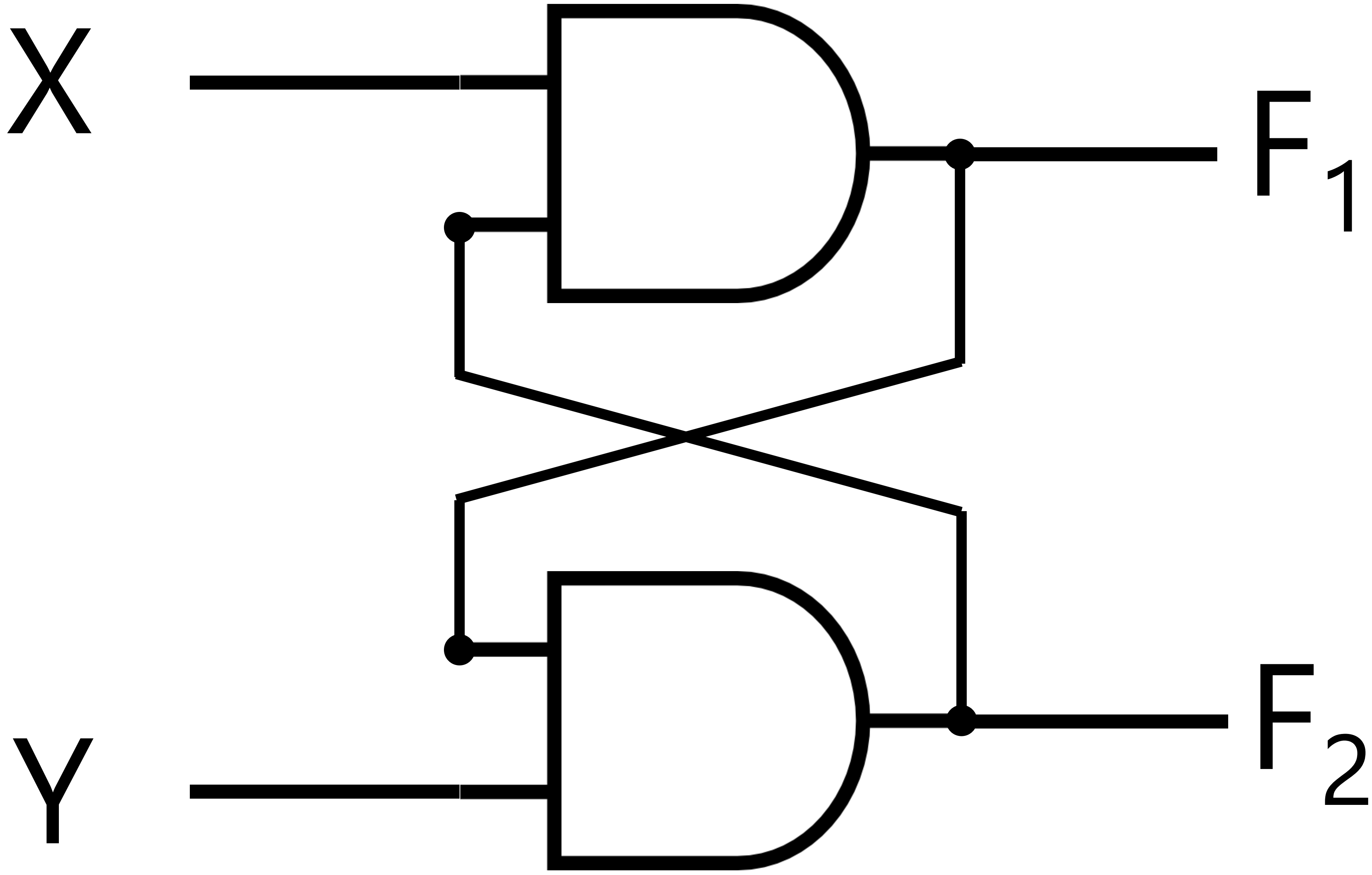
We have two inputs: X and X_{T-1}

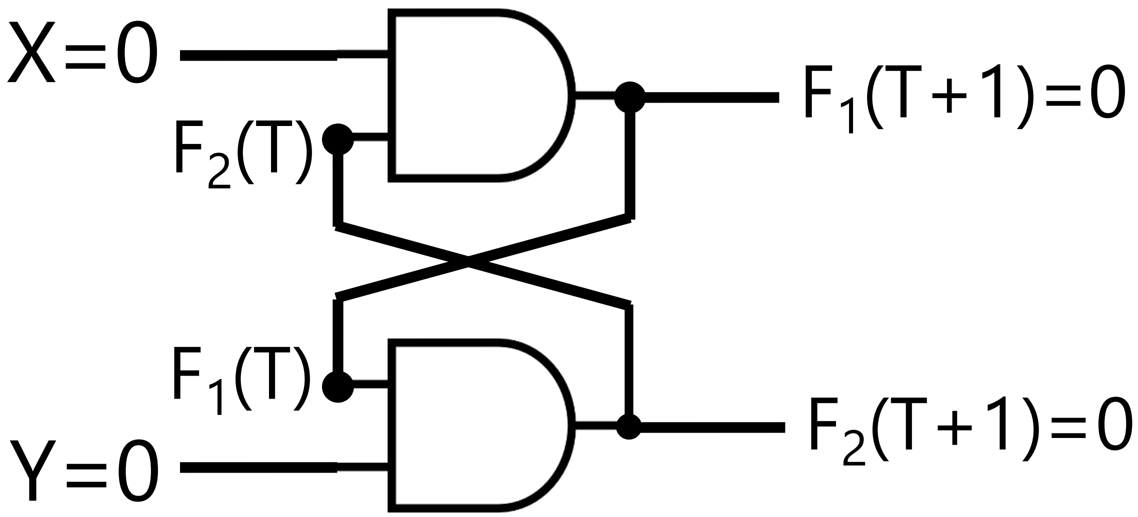


This design is logically correct!
But what’s the problem?

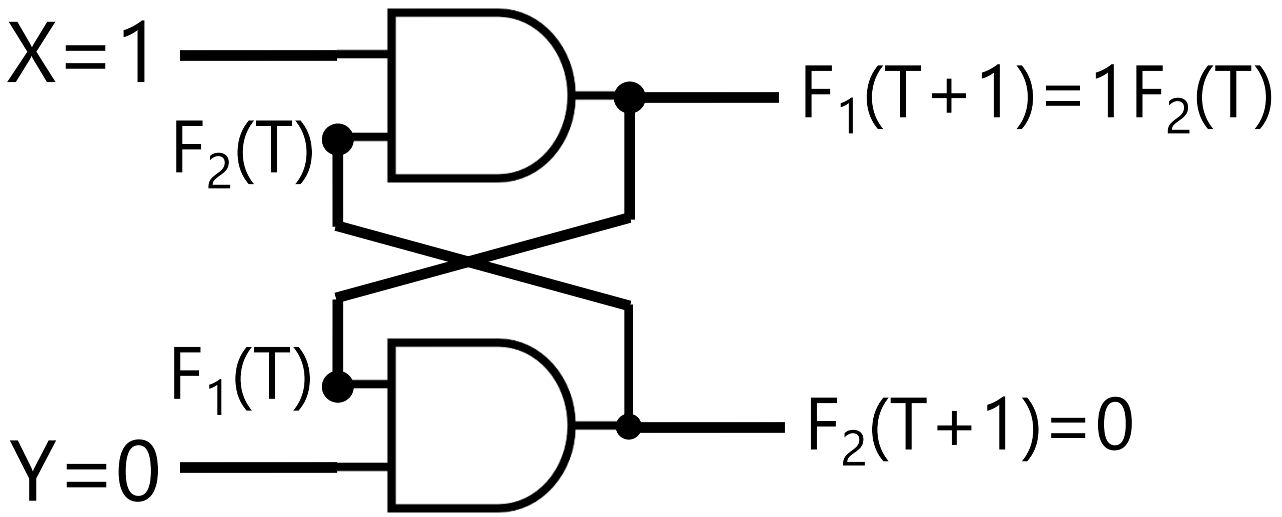


This design is also logically correct.
But what’s the problem?

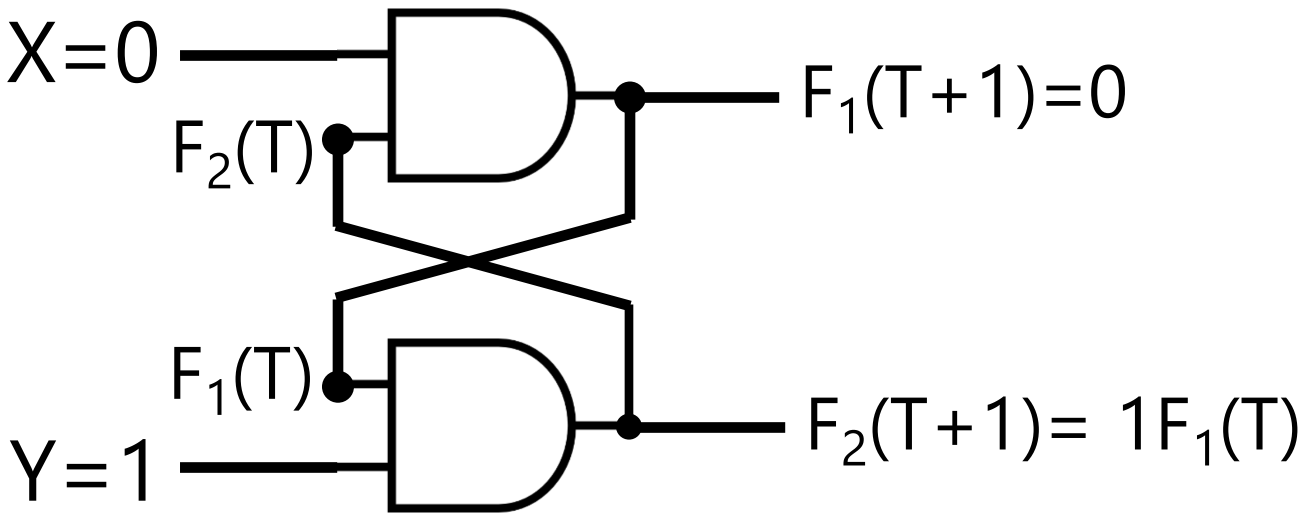

X
Y
F1
F2



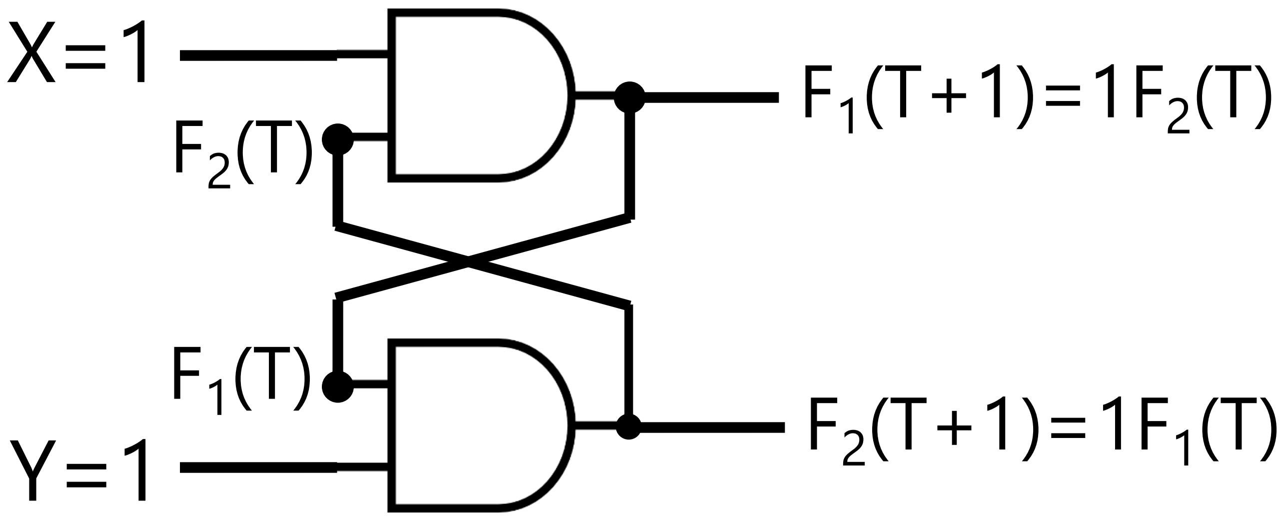
Y	X	$F_1(T+1)$	$F_2(T+1)$
0	0	0	0
0	1		
1	0		
1	1		



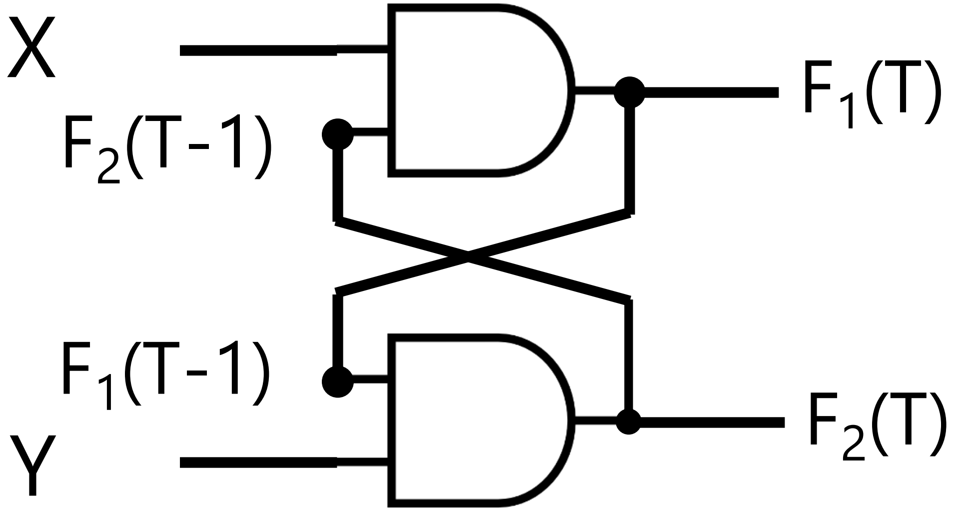
Y	X	$F_1(T+1)$	$F_2(T+1)$
0	0	0	0
0	1	$F_2(T)$	0
1	0		
1	1		



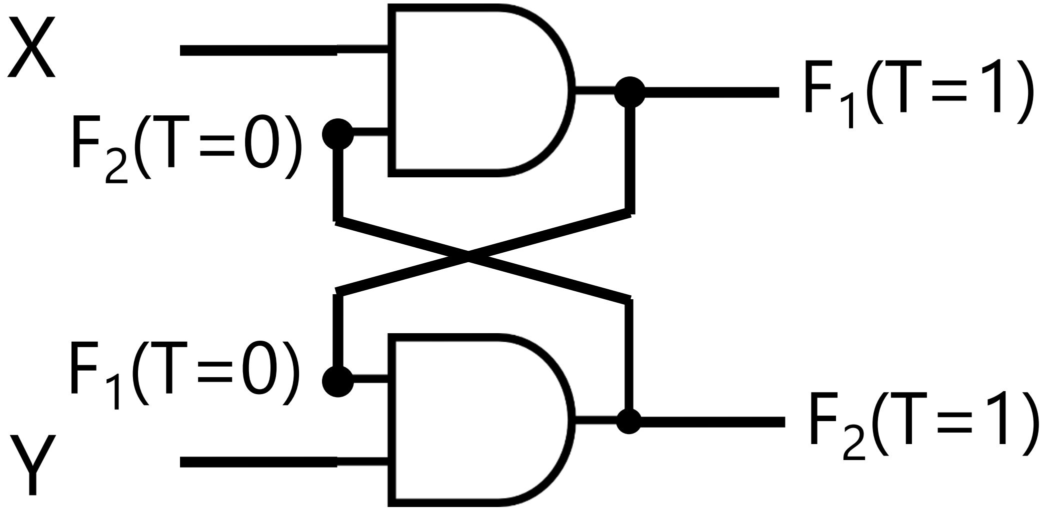
Y	X	$F_1(T+1)$	$F_2(T+1)$
0	0	0	0
0	1	$F_2(T)$	0
1	0	0	$F_1(T)$
1	1		



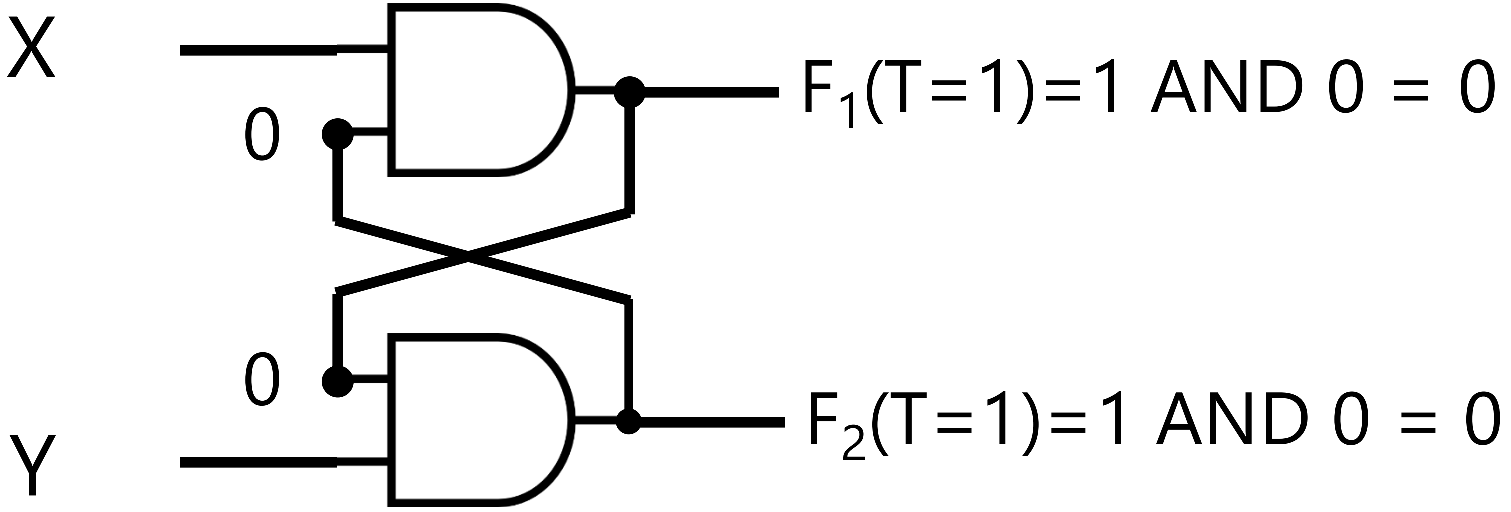
Y	X	$F_1(T+1)$	$F_2(T+1)$
0	0	0	0
0	1	$F_2(T)$	0
1	0	0	$F_1(T)$
1	1	$F_2(T)$	$F_1(T)$



Y	X	$F_1(T)$	$F_2(T)$
0	0	0	0
0	1	$F_2(T-1)$	0
1	0	0	$F_1(T-1)$
1	1	$F_2(T-1)$	$F_1(T-1)$

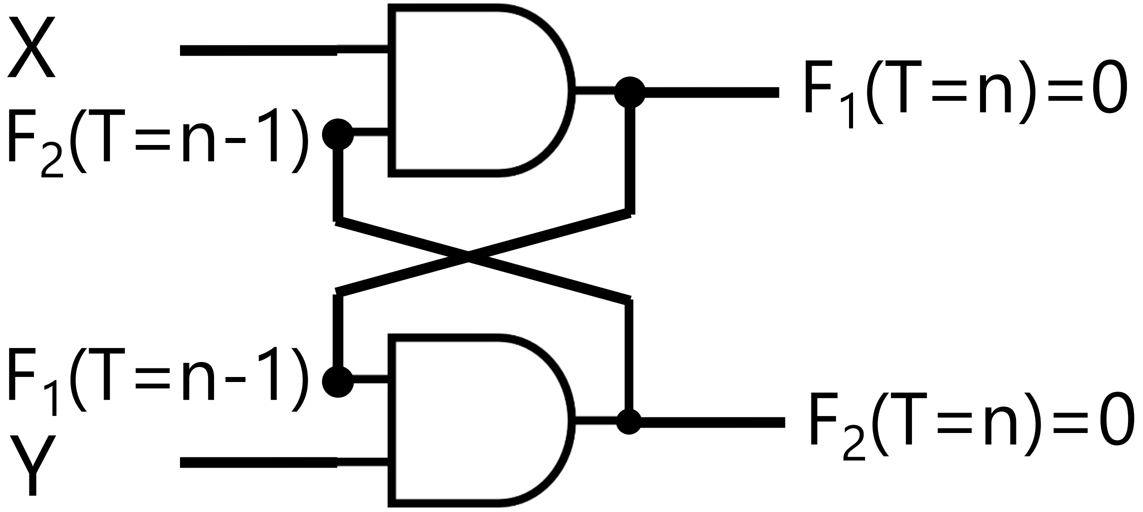


Y	X	$F_1(T=1)$	$F_2(T=1)$
0	0	0	0
0	1	$F_2(T=0)$	0
1	0	0	$F_1(T=0)$
1	1	$F_2(T=0)$	$F_1(T=0)$



by default, in positive logic $F(T=0) = 0$

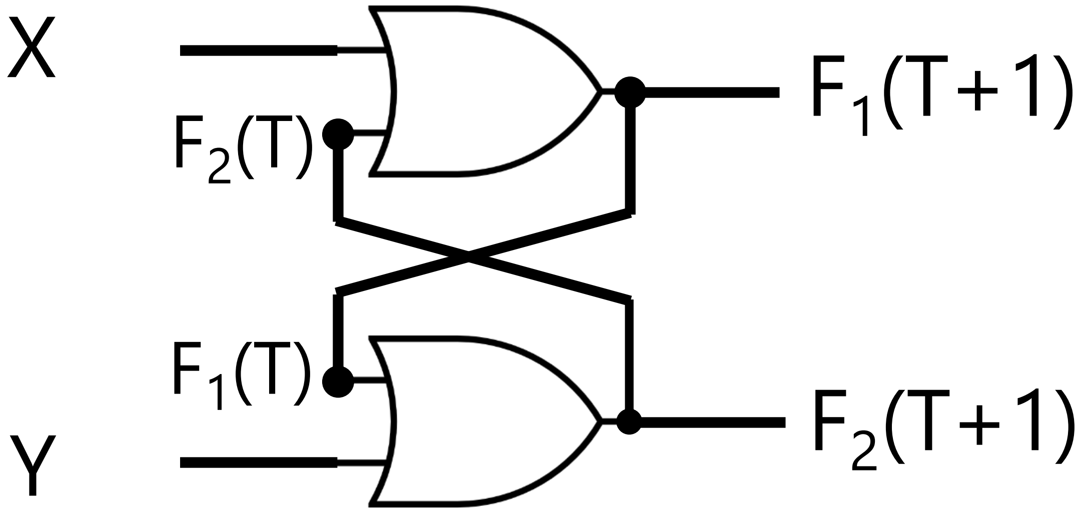
Y	X	$F_1(T=1)$	$F_2(T=1)$
0	0	0	0
0	1	0	0
1	0	0	0
1	1	0	0



When it goes to 00 state, never recover to 1!

Y	X	$F_1(T=n)$	$F_2(T=n)$
0	0	0	0
0	1	0	0
1	0	0	0
1	1	0	0

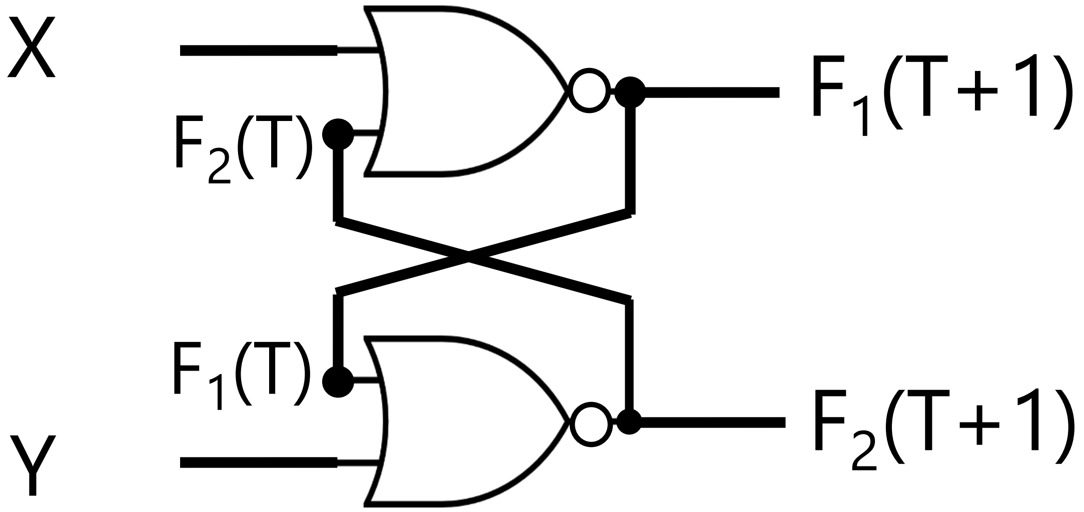
Try OR gates



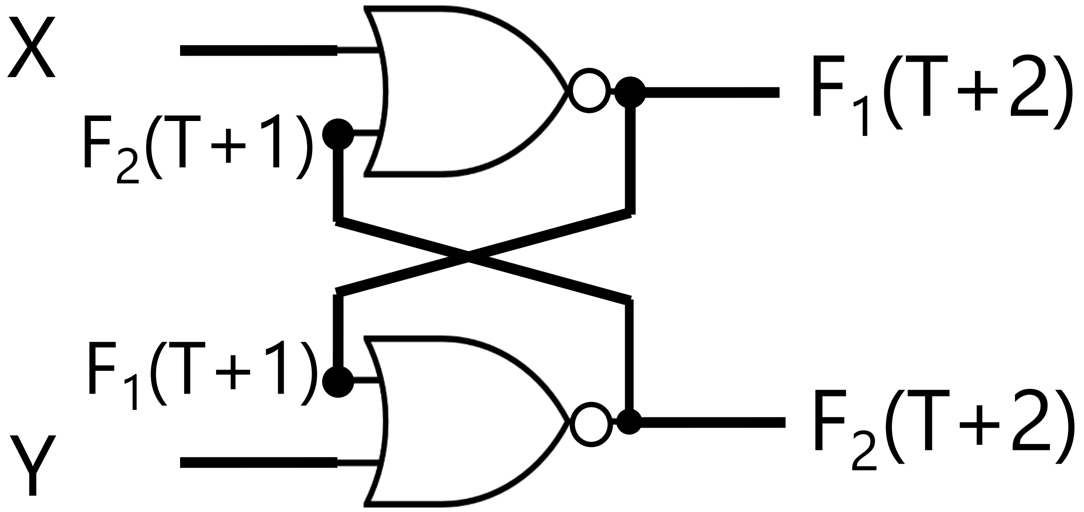
How about OR gates? Do they solve the problem? No! Why?

Y	X	$F_1(T+1)$	$F_2(T+1)$
0	0		
0	1		
1	0		
1	1		

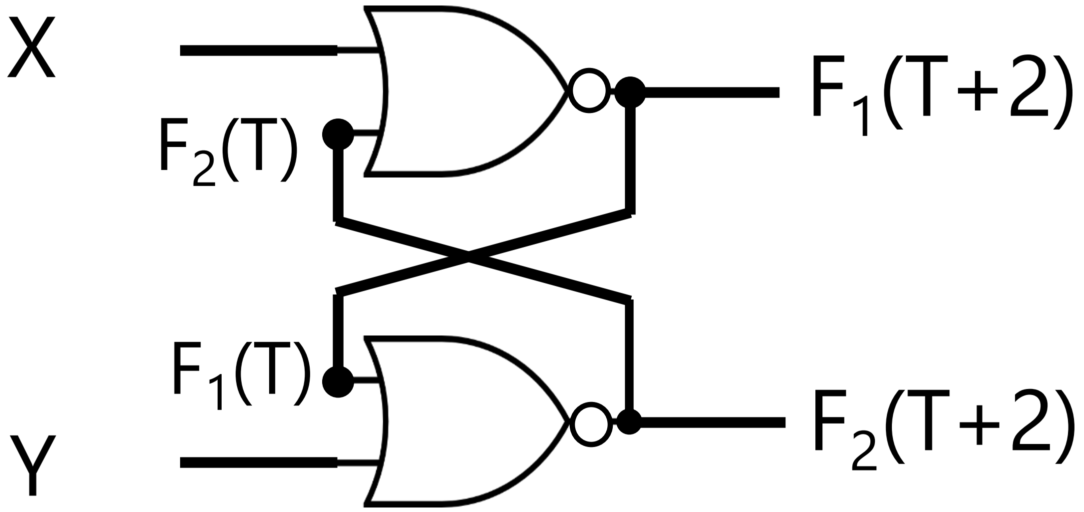
Try NOR gates



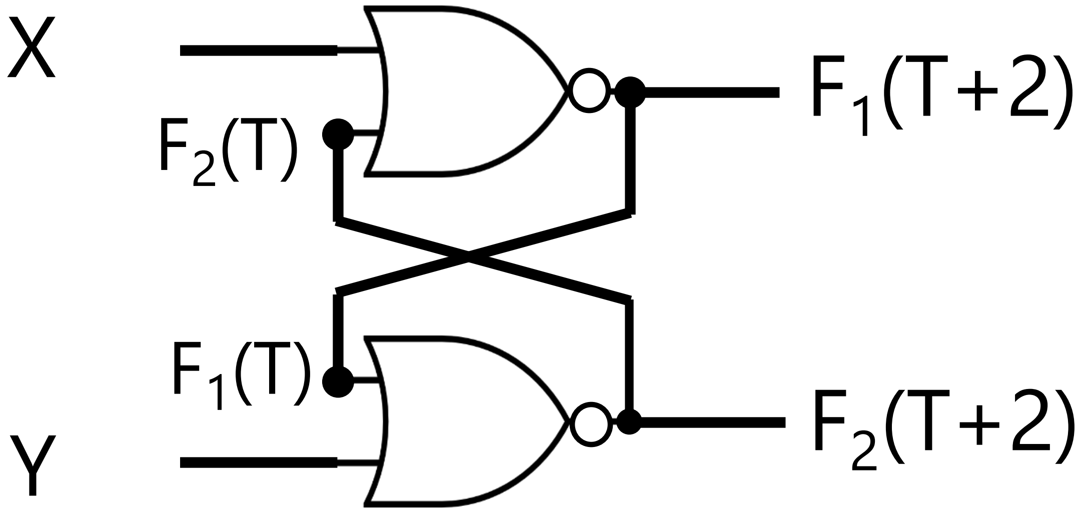
Y	X	$F_1(T+1)$	$F_2(T+1)$
0	0	$F'_2(T)$	$F'_1(T)$
0	1	0	$F'_1(T)$
1	0	$F'_2(T)$	0
1	1	0	0



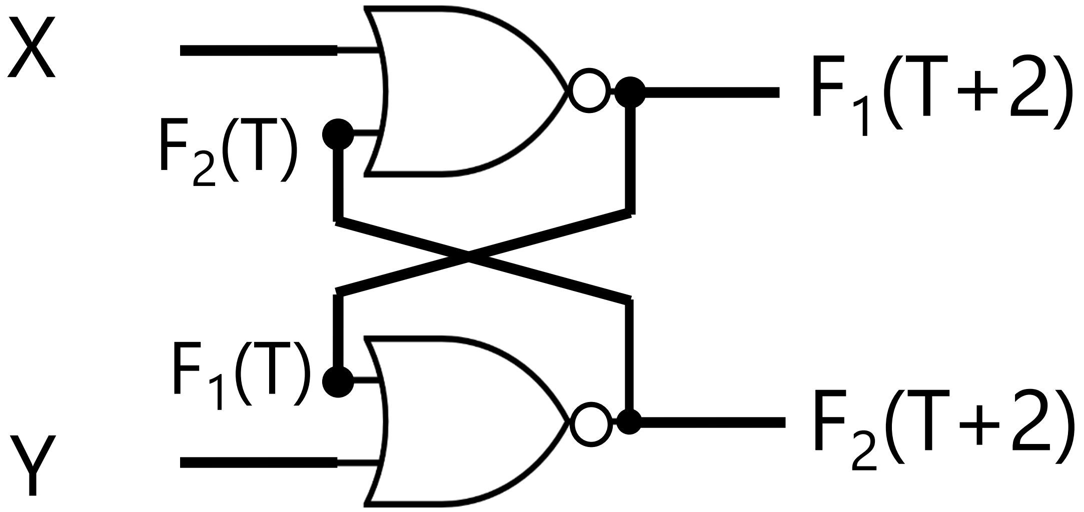
Y	X	$F_1(T+1)$	$F_2(T+1)$	$F_1(T+2)$	$F_2(T+2)$
0	0	$F'_2(T)$	$F'_1(T)$	$F'_2(T+1)$	$F'_1(T+1)$
0	1	0	$F'_1(T)$	0	$F'_1(T+1)$
1	0	$F'_2(T)$	0	$F'_2(T+1)$	0
1	1	0	0	0	0



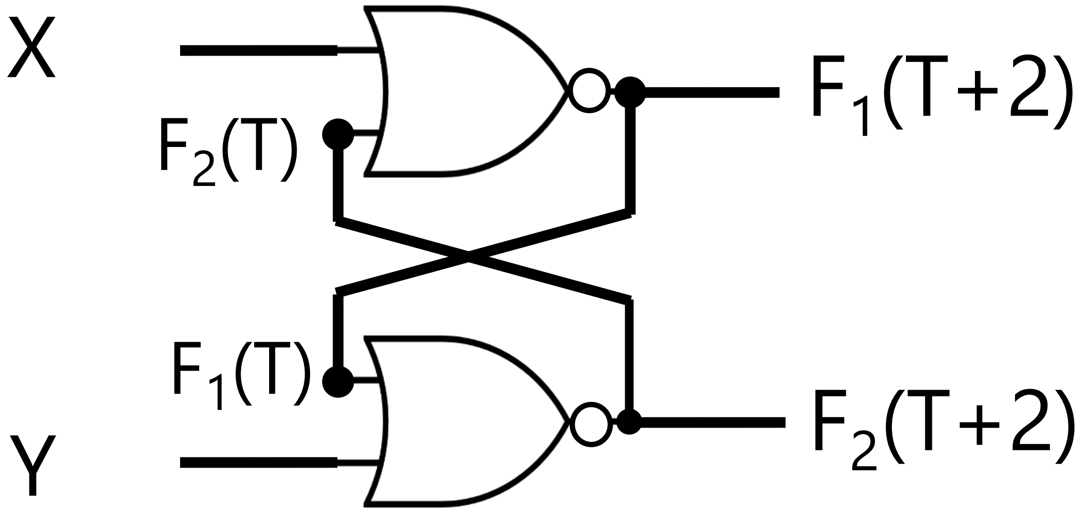
Y	X	$F_1(T+1)$	$F_2(T+1)$	$F_1(T+2)$	$F_2(T+2)$
0	0	$F'_2(T)$	$F'_1(T)$	$F'_2(T+1)$	$F'_1(T+1)$
0	1	0	$F'_1(T)$	0	$F'_1(T+1)$
1	0	$F'_2(T)$	0	$F'_2(T+1)$	0
1	1	0	0	0	0



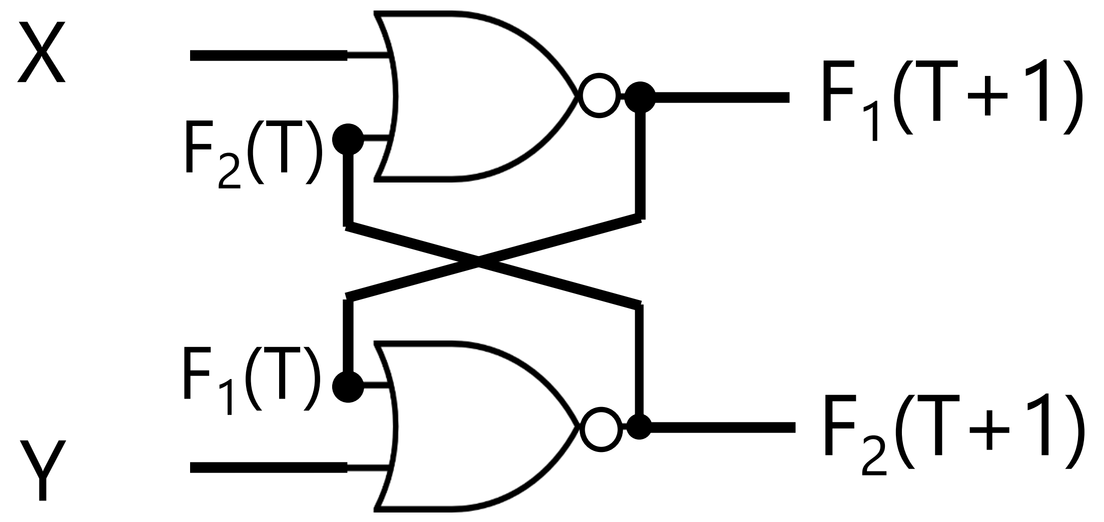
Y	X	$F_1(T+1)$	$F_2(T+1)$	$F_1(T+2)$	$F_2(T+2)$
0	0	$F'_2(T)$	$F'_1(T)$	$F_1(T)$	$F'_1(T+1)$
0	1	0	$F'_1(T)$	0	$F'_1(T+1)$
1	0	$F'_2(T)$	0	$F'_2(T+1)$	0
1	1	0	0	0	0



Y	X	$F_1(T+1)$	$F_2(T+1)$	$F_1(T+2)$	$F_2(T+2)$
0	0	$F'_2(T)$	$F'_1(T)$	$F_1(T)$	$F_2(T)$
0	1	0	$F'_1(T)$	0	$F'_1(T+1)$
1	0	$F'_2(T)$	0	$F'_2(T+1)$	0
1	1	0	0	0	0

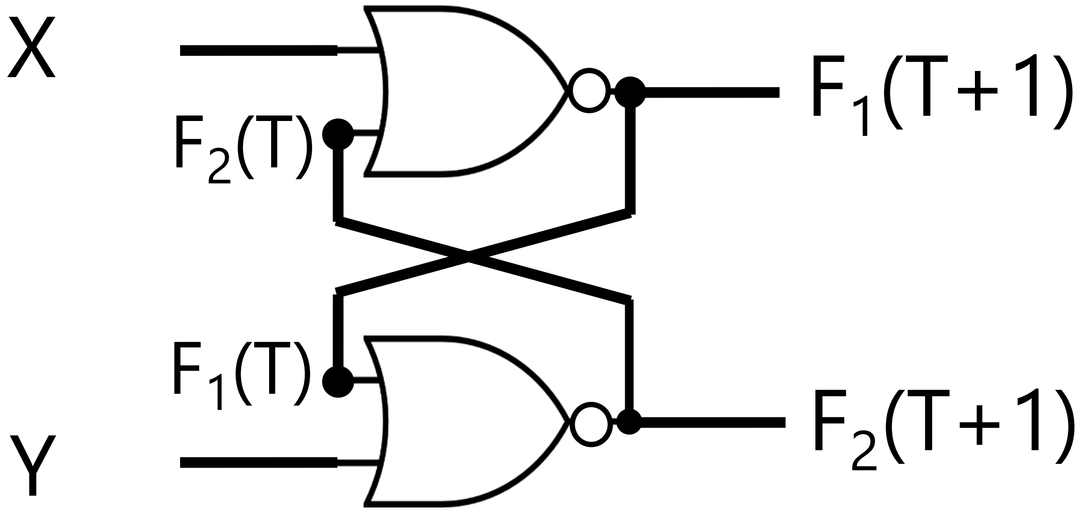


Y	X	$F_1(T+1)$	$F_2(T+1)$	$F_1(T+2)$	$F_2(T+2)$
0	0	$F'_2(T)$	$F'_1(T)$	$F_1(T)$	$F_2(T)$
0	1	0	$F'_1(T)$	0	1
1	0	$F'_2(T)$	0	$F'_2(T+1)$	0
1	1	0	0	0	0

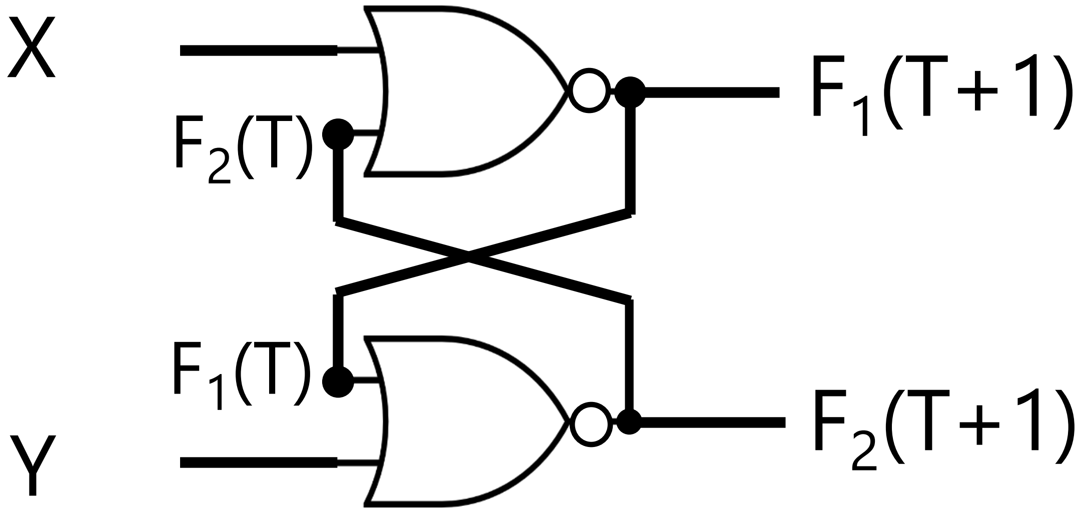


WOW! If wait longer, it stores the current state!

Y	X	$F_1(T+1)$	$F_2(T+1)$	$F_1(T+2)$	$F_2(T+2)$
0	0	$F'_2(T)$	$F'_1(T)$	$F_1(T)$	$F_2(T)$
0	1	0	$F'_1(T)$	0	1
1	0	$F'_2(T)$	0	1	0
1	1	0	0	0	0

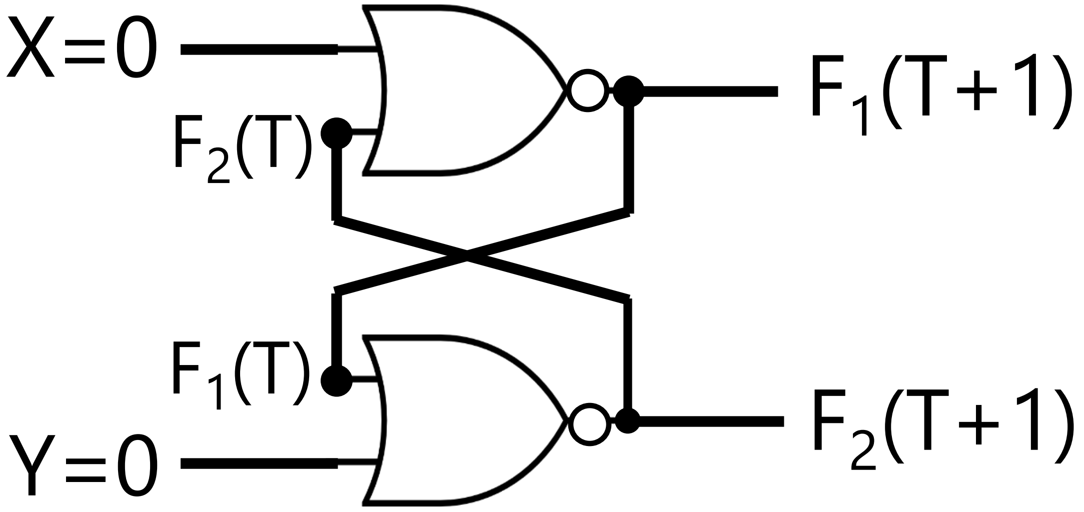


Y	X	$F_1(T+1)$	$F_2(T+1)$	$F_1(T+2)$	$F_2(T+2)$	$F_1(T+3)$	$F_2(T+3)$	$F_1(T+4)$	$F_2(T+4)$
0	0	$F'_2(T)$	$F'_1(T)$	$F_1(T)$	$F_2(T)$	$F'_2(T)$	$F'_1(T)$	$F'_2(T+3)$	$F'_1(T+3)$
0	1	0	$F'_1(T)$	0	1	0	1	0	$F'_1(T+3)$
1	0	$F'_2(T)$	0	1	0	0	0	$F'_2(T+3)$	0
1	1	0	0	0	0	0	0	0	0



Y	X	$F_1(T+1)$	$F_2(T+1)$	$F_1(T+2)$	$F_2(T+2)$	$F_1(T+3)$	$F_2(T+3)$	$F_1(T+4)$	$F_2(T+4)$
0	0	$F'_2(T)$	$F'_1(T)$	$F_1(T)$	$F_2(T)$	$F'_2(T)$	$F'_1(T)$	$F_1(T)$	$F_2(T)$
0	1	0	$F'_1(T)$	0	1	0	1	0	1
1	0	$F'_2(T)$	0	1	0	0	0	1	0
1	1	0	0	0	0	0	0	0	0

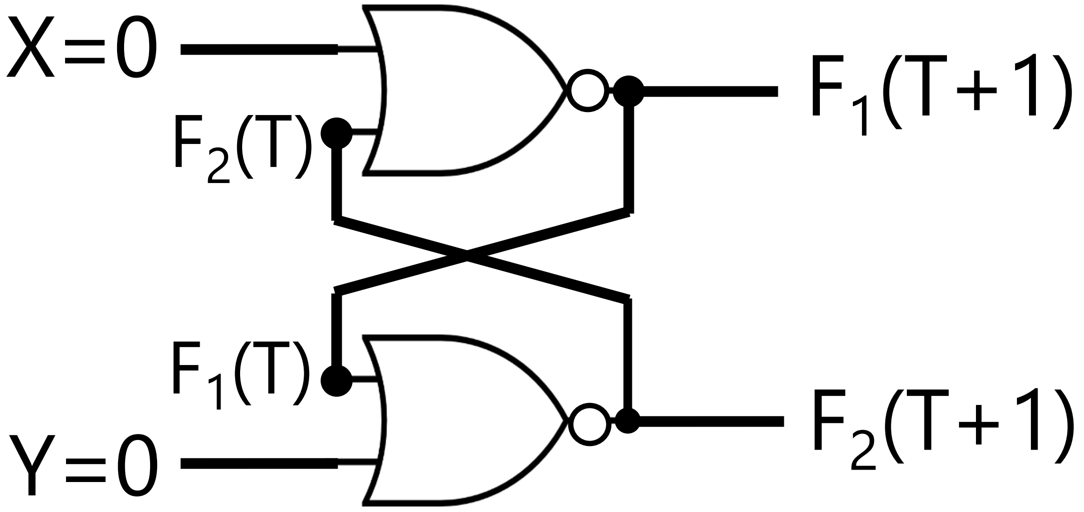
Store/Hold Action



Y	X	$F_1(T+1)$	$F_2(T+1)$	$F_1(T+2)$	$F_2(T+2)$	$F_1(T+3)$	$F_2(T+3)$	$F_1(T+4)$	$F_2(T+4)$
0	0	$F'_2(T)$	$F'_1(T)$	$F_1(T)$	$F_2(T)$	$F'_2(T)$	$F'_1(T)$	$F_1(T)$	$F_2(T)$
0	1	0	$F'_1(T)$	0	1	0	1	0	1
1	0	$F'_2(T)$	0	1	0	0	0	1	0
1	1	0	0	0	0	0	0	0	0

Store/Hold Action

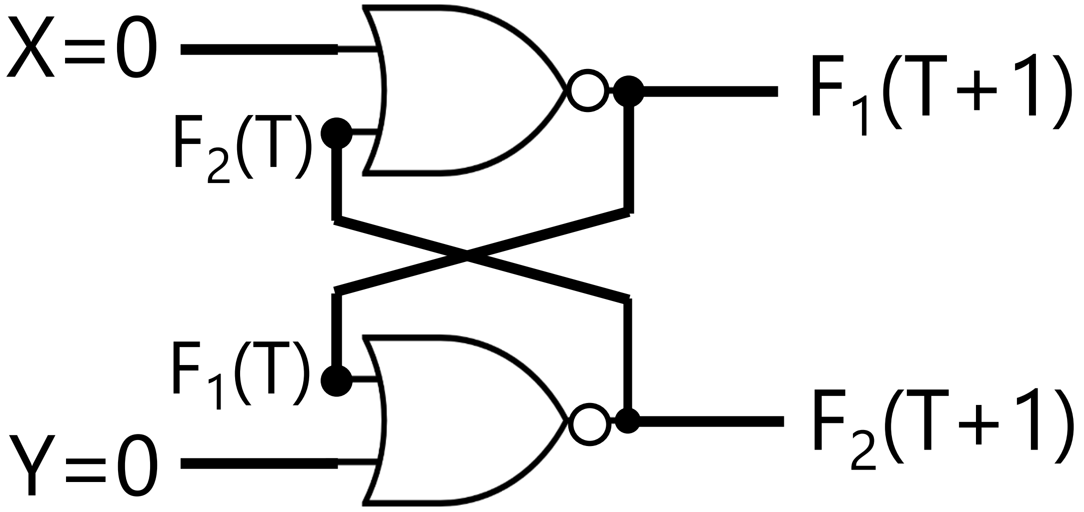
2T → T



Y	X	$F_1(T+2)$	$F_2(T+2)$
0	0	$F_1(T)$	$F_2(T)$
0	1	0	1
1	0	1	0
1	1	0	0

Store/Hold Action

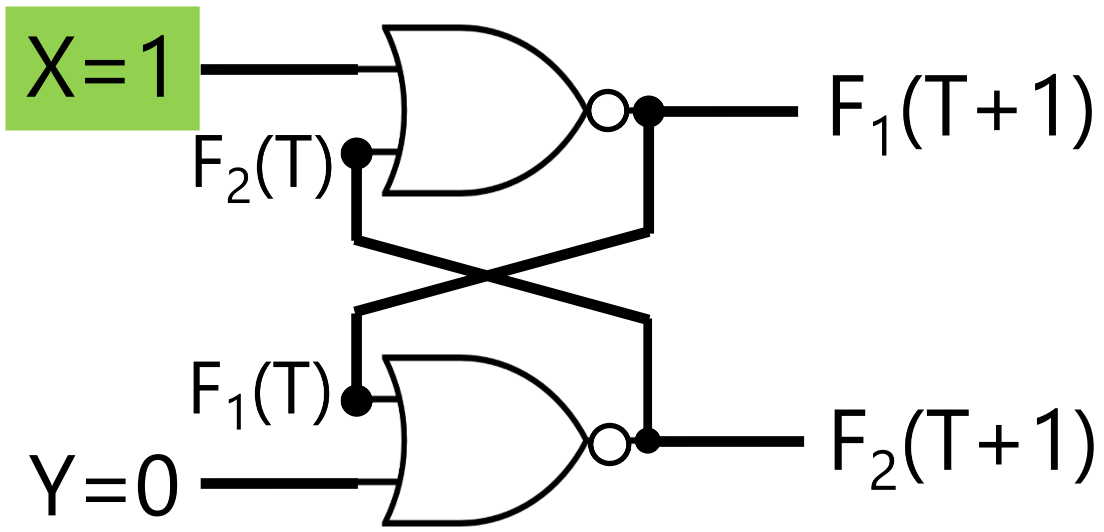
$2T \rightarrow T$



Y	X	$F_1(T+1)$	$F_2(T+1)$
0	0	$F_1(T)$	$F_2(T)$
0	1	0	1
1	0	1	0
1	1	0	0

Reset Action

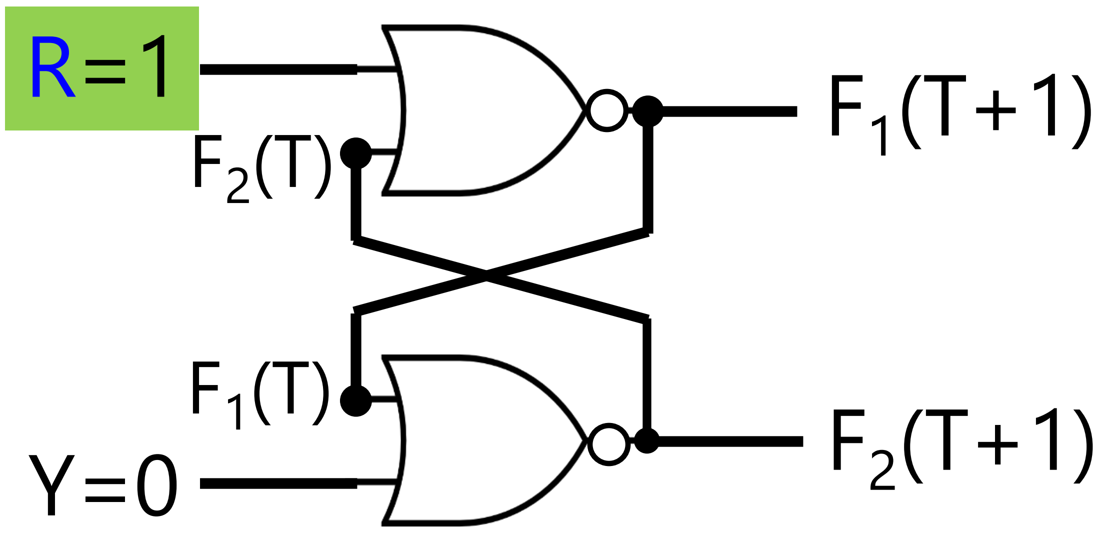
$2T \rightarrow T$



Y	X	$F_1(T+1)$	$F_2(T+1)$
0	0	$F_1(T)$	$F_2(T)$
0	1	0	1
1	0	1	0
1	1	0	0

Reset Action

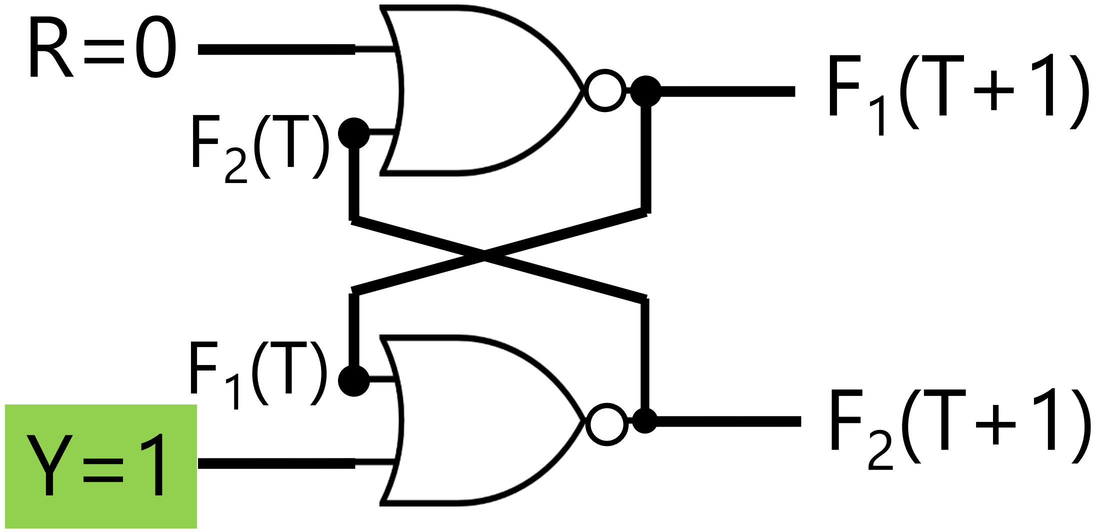
$2T \rightarrow T$



Y	R	$F_1(T+1)$	$F_2(T+1)$
0	0	$F_1(T)$	$F_2(T)$
0	1	0	1
1	0	1	0
1	1	0	0

Set Action

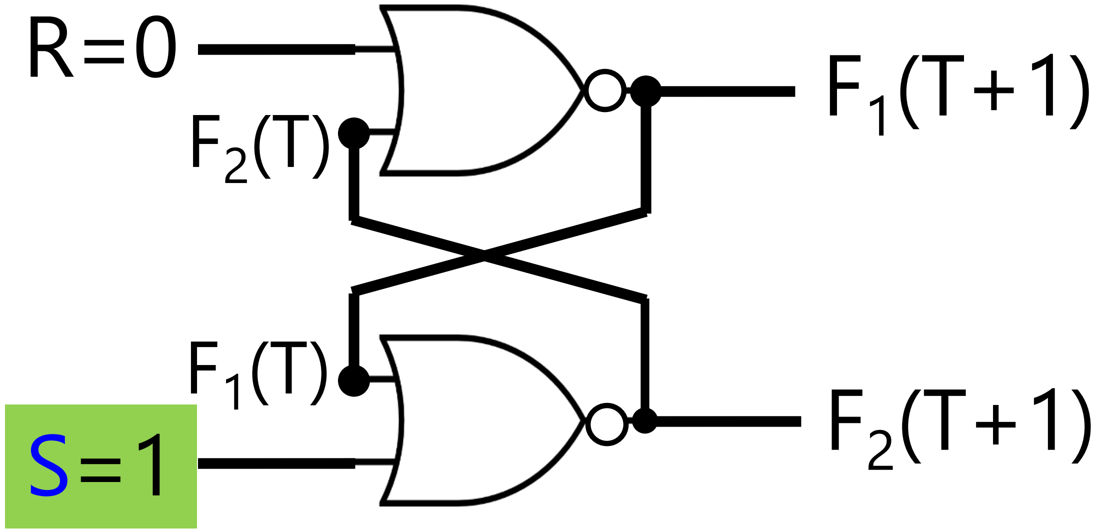
$2T \rightarrow T$



Y	R	$F_1(T+1)$	$F_2(T+1)$
0	0	$F_1(T)$	$F_2(T)$
0	1	0	1
1	0	1	0
1	1	0	0

Set Action

$2T \to T$

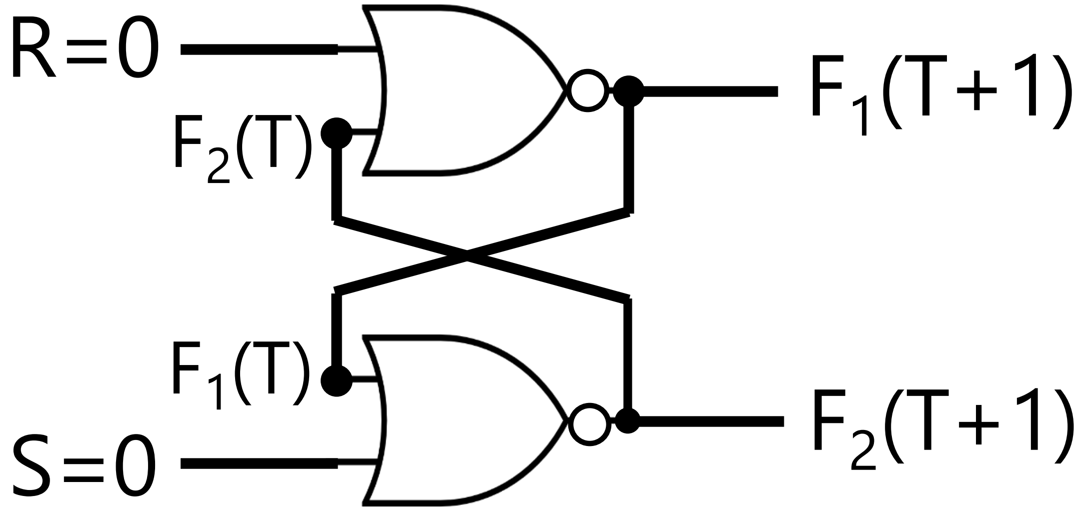


S	R	$F_1(T+1)$	$F_2(T+1)$
0	0	$F_1(T)$	$F_2(T)$
0	1	0	1
1	0	1	0
1	1	0	0

Store/Hold Action

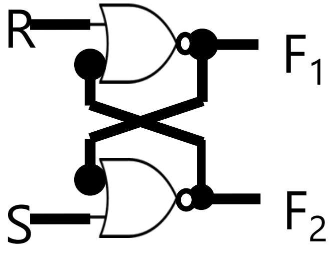
2T → T

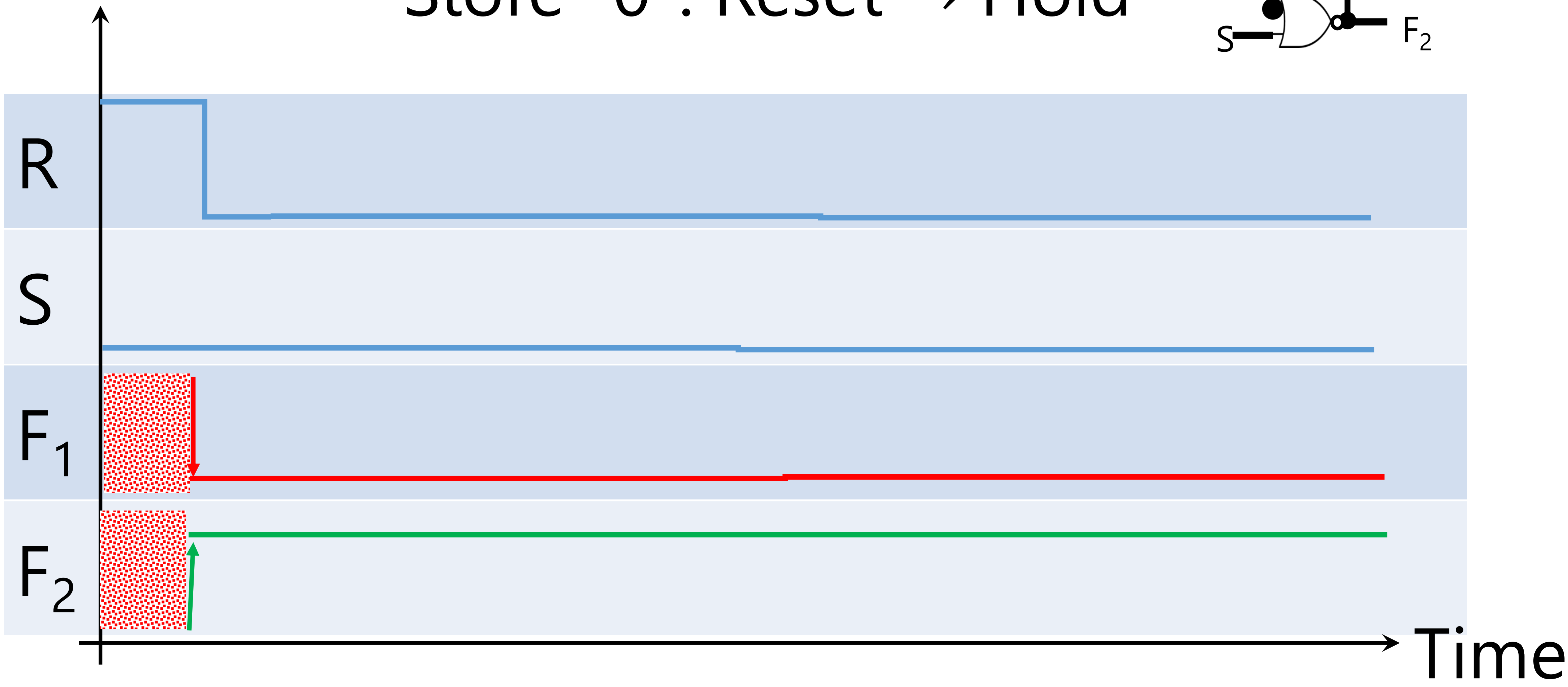
No Set & No Reset



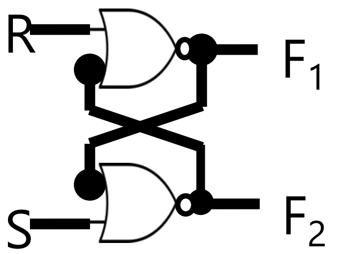
S	R	$F_1(T+1)$	$F_2(T+1)$
0	0	$F_1(T)$	$F_2(T)$
0	1	0	1
1	0	1	0
1	1	0	0

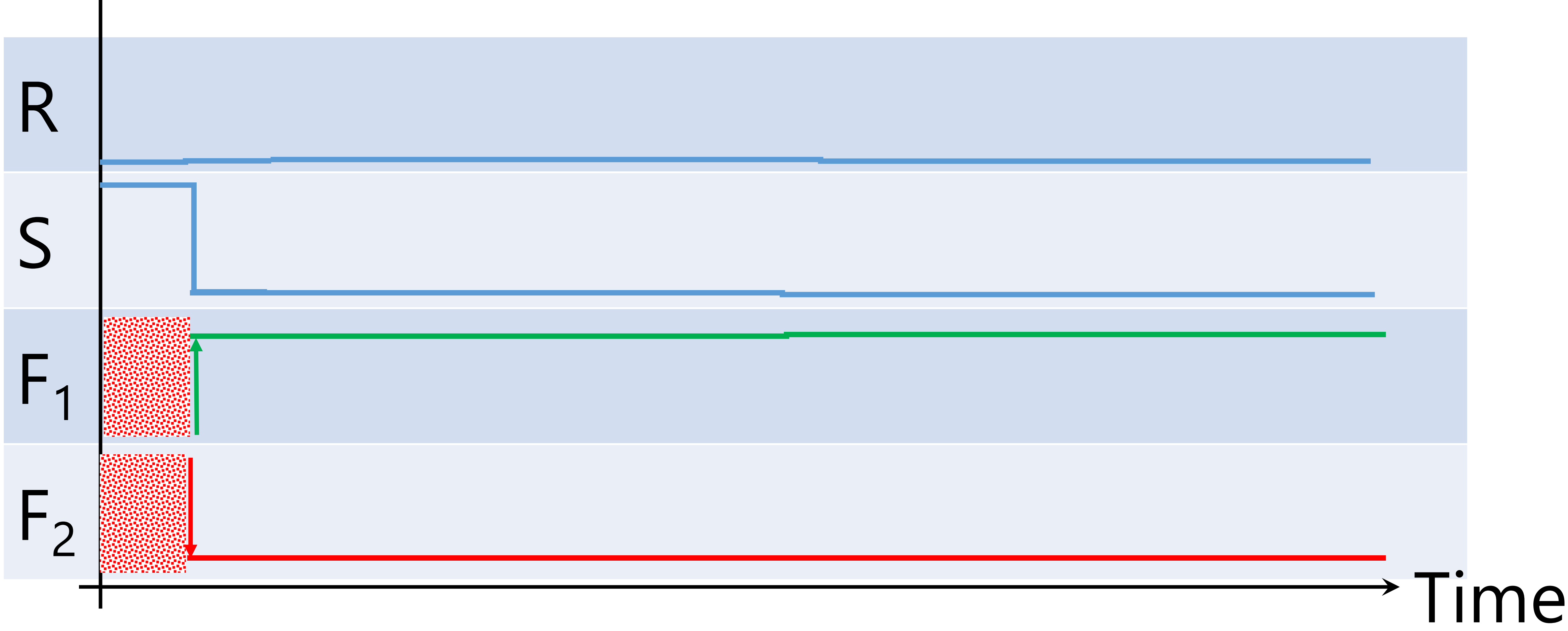
Store “0”: Reset → Hold

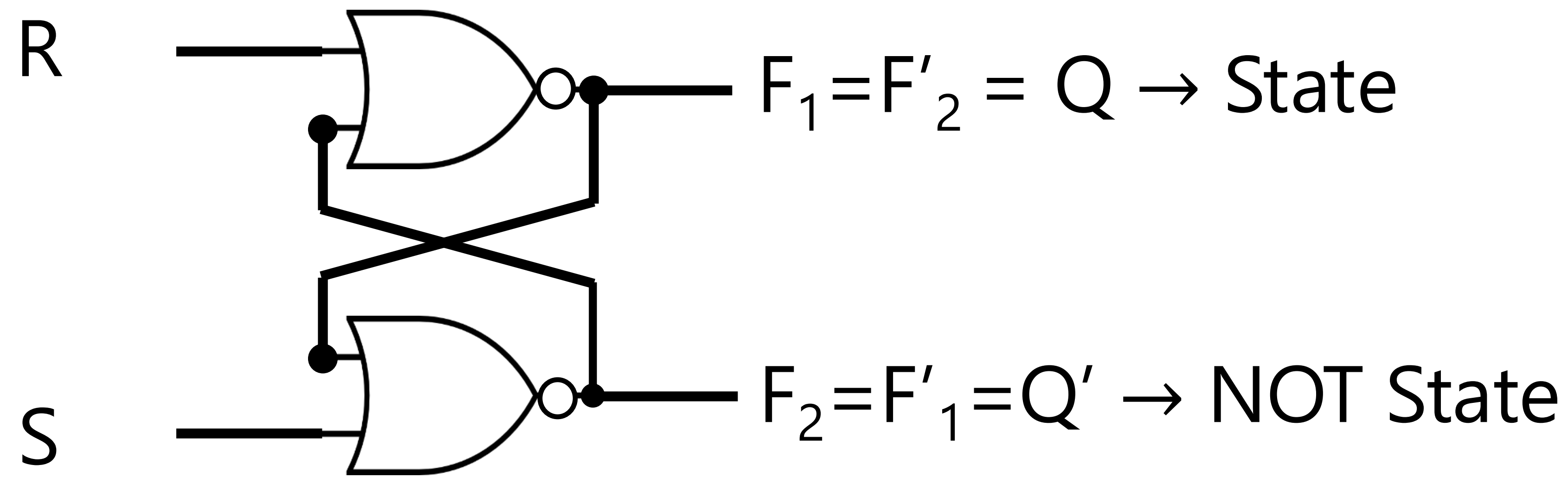




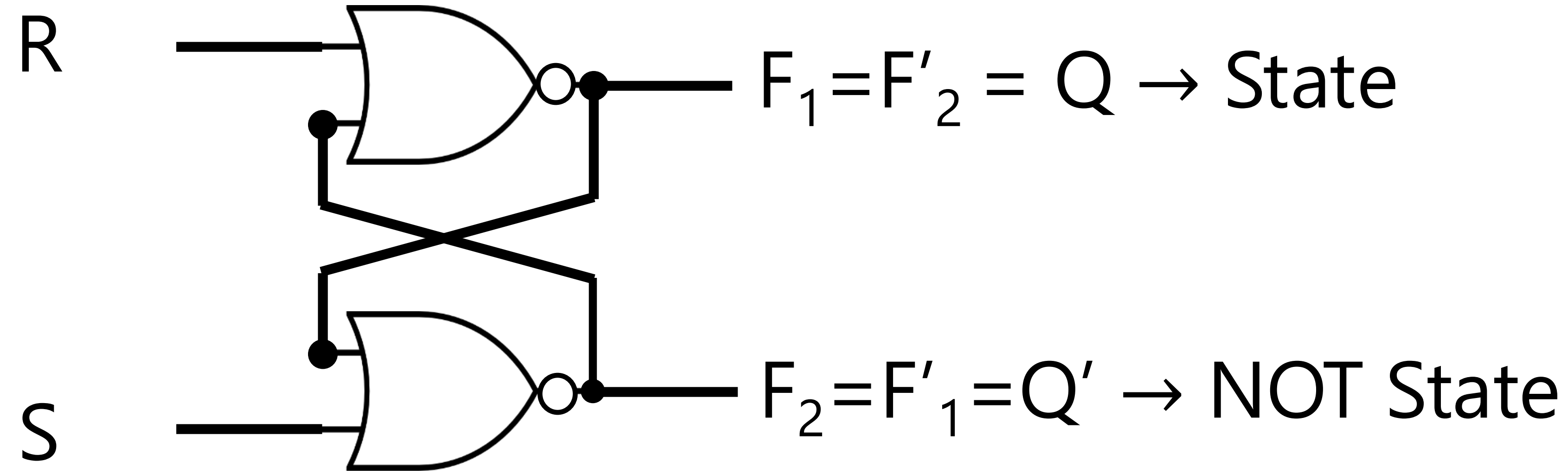
Store “1”: Set → Hold







S	R	Q	Q′
0	0	Q_t	Q'_t
0	1	0	1
1	0	1	0
1	1	0	0

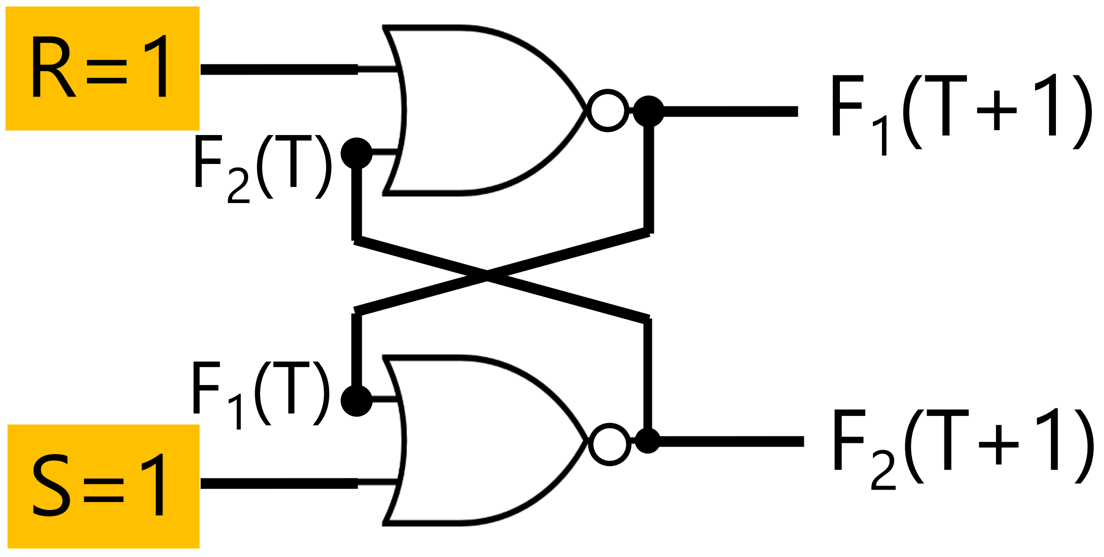


S	R	Q	Q'
0	0	Q_t	Q'_t
0	1	0	1
1	0	1	0
1	1	0	0

Forbidden Action

$2T \rightarrow T$

~~*Set & Reset*~~

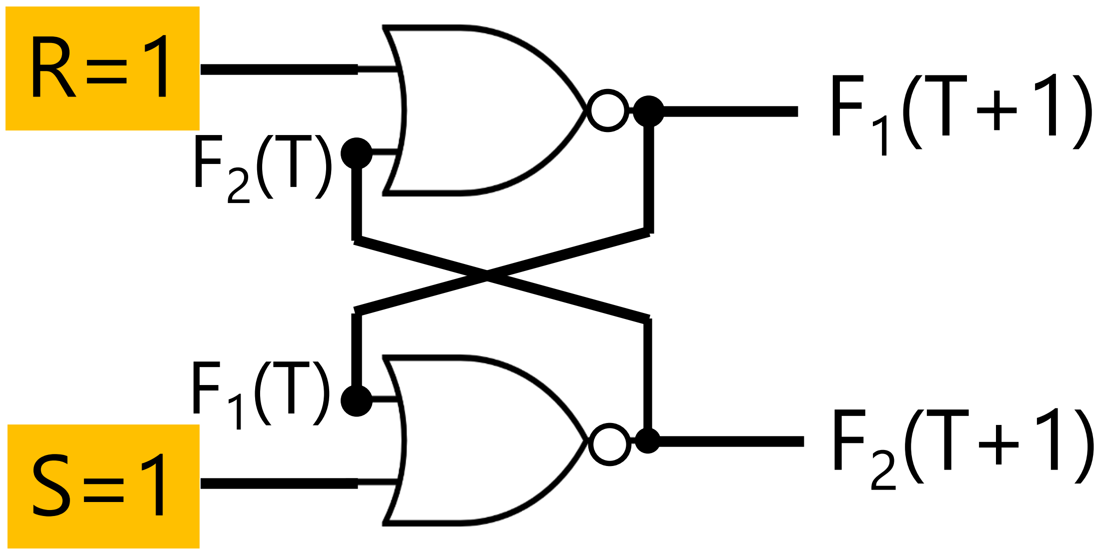


S	R	$F_1(T+1)$	$F_2(T+1)$
0	0	$F_1(T)$	$F_2(T)$
0	1	0	1
1	0	1	0
1	1	0	0

Forbidden Action

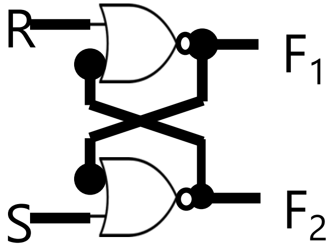
$2T \rightarrow T$

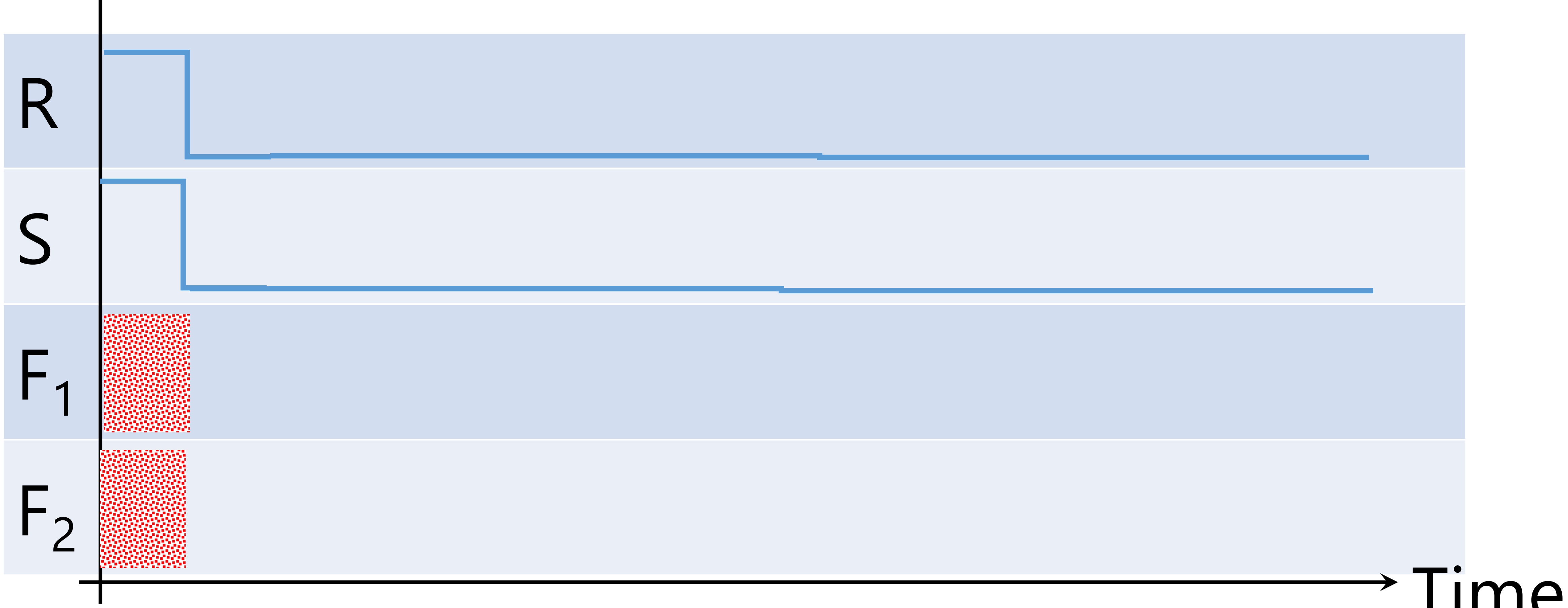
~~*Set & Reset*~~



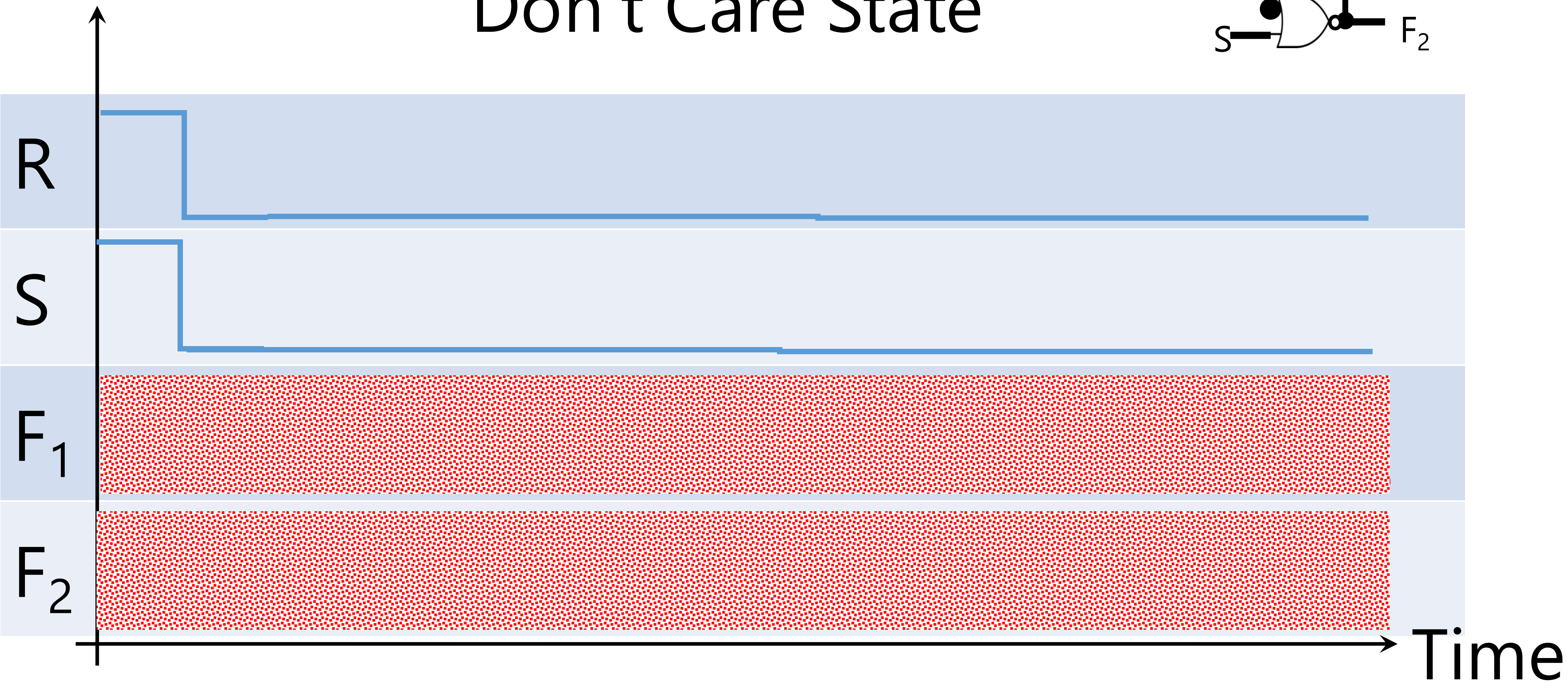
S	R	$F_1(T+1)$	$F_2(T+1)$
0	0	$F_1(T)$	$F_2(T)$
0	1	0	1
1	0	1	0
1	1	X	X

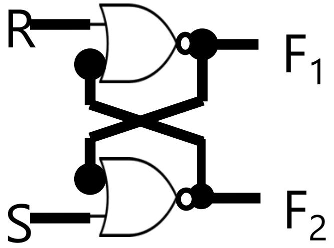
Don’t Care State

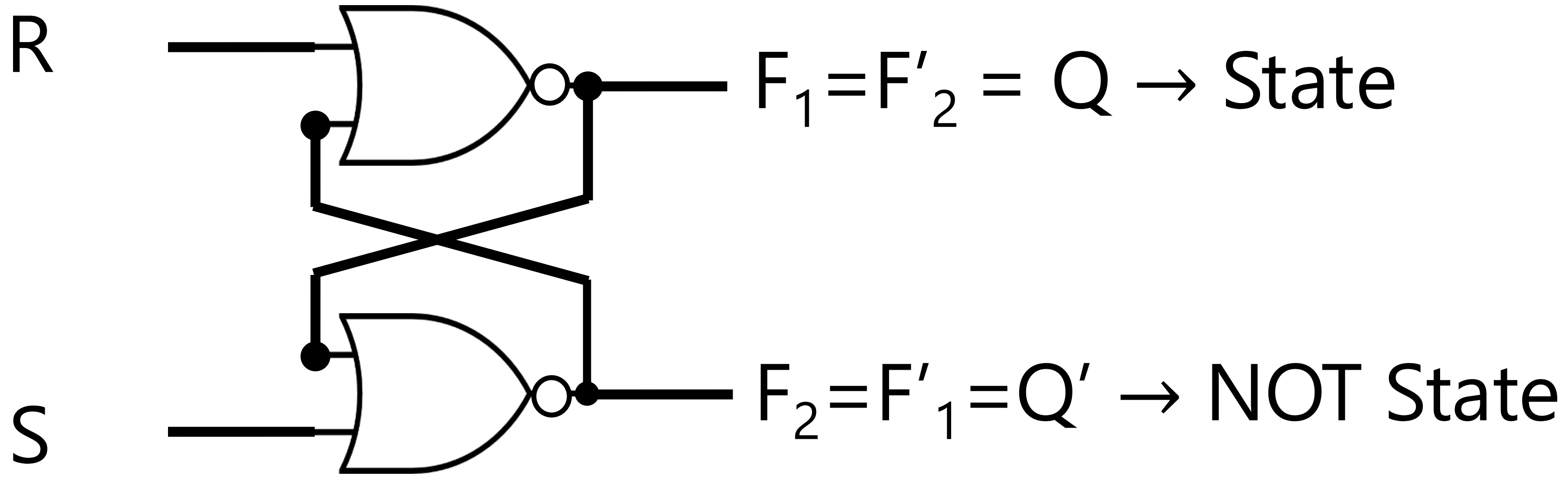




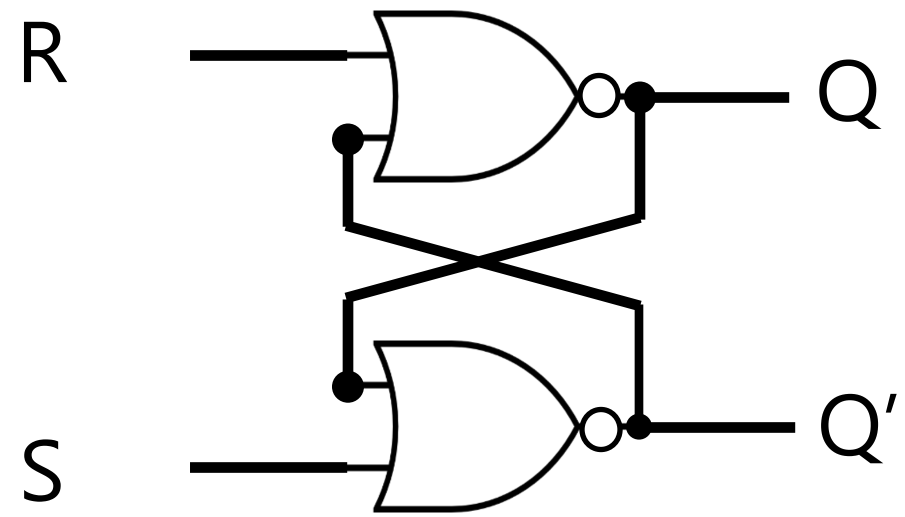
Don't Care State



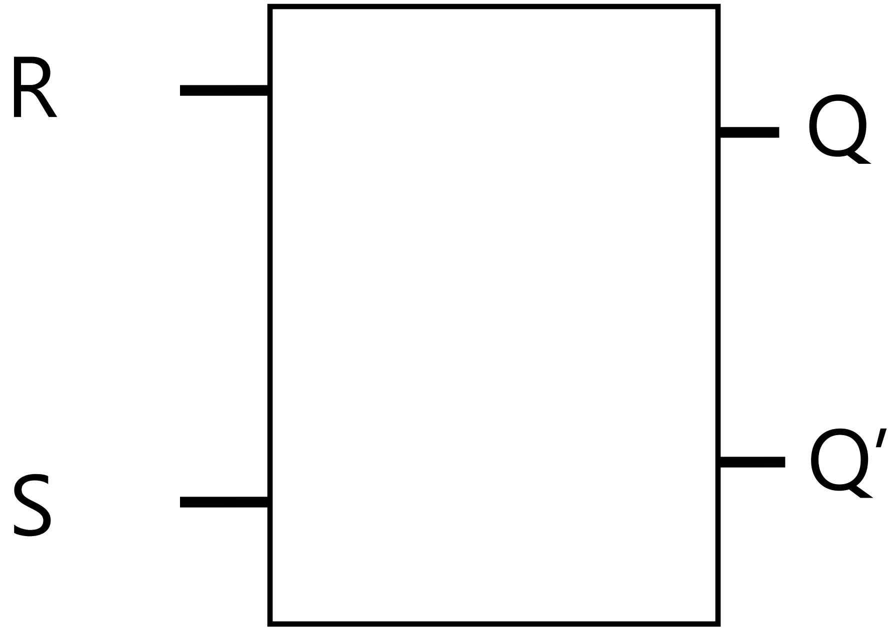




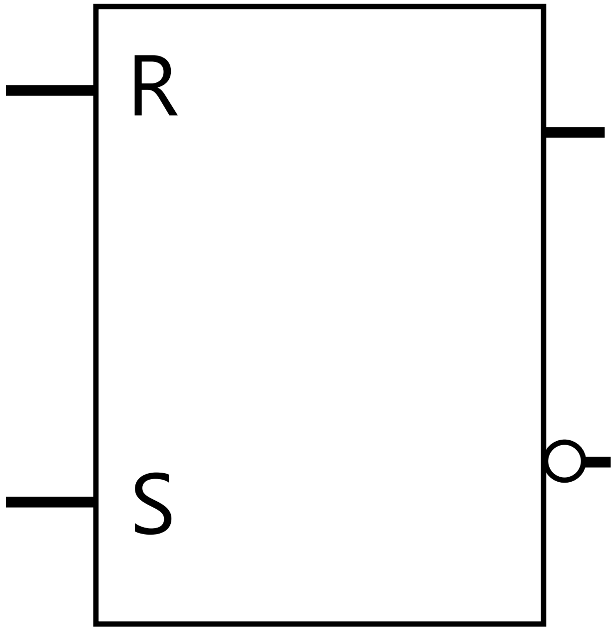
S	R	Q	Q′
0	0	Q_t	Q'_t
0	1	0	1
1	0	1	0
1	1	x	x



	S	R	Q	Q′
Hold	0	0	Q_t	Q'_t
Reset	0	1	0	1
Set	1	0	1	0
	1	1	x	x



	S	R	Q	Q′
Hold	0	0	Q_t	Q'_t
Reset	0	1	0	1
Set	1	0	1	0
	1	1	X	X

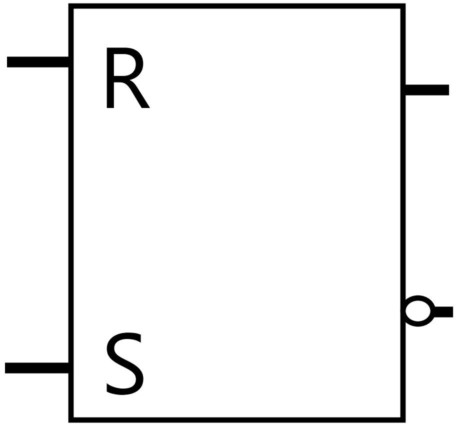


	S	R	Q	Q′
Hold	0	0	Q_t	Q'_t
Reset	0	1	0	1
Set	1	0	1	0
	1	1	x	x

State Transition Diagram

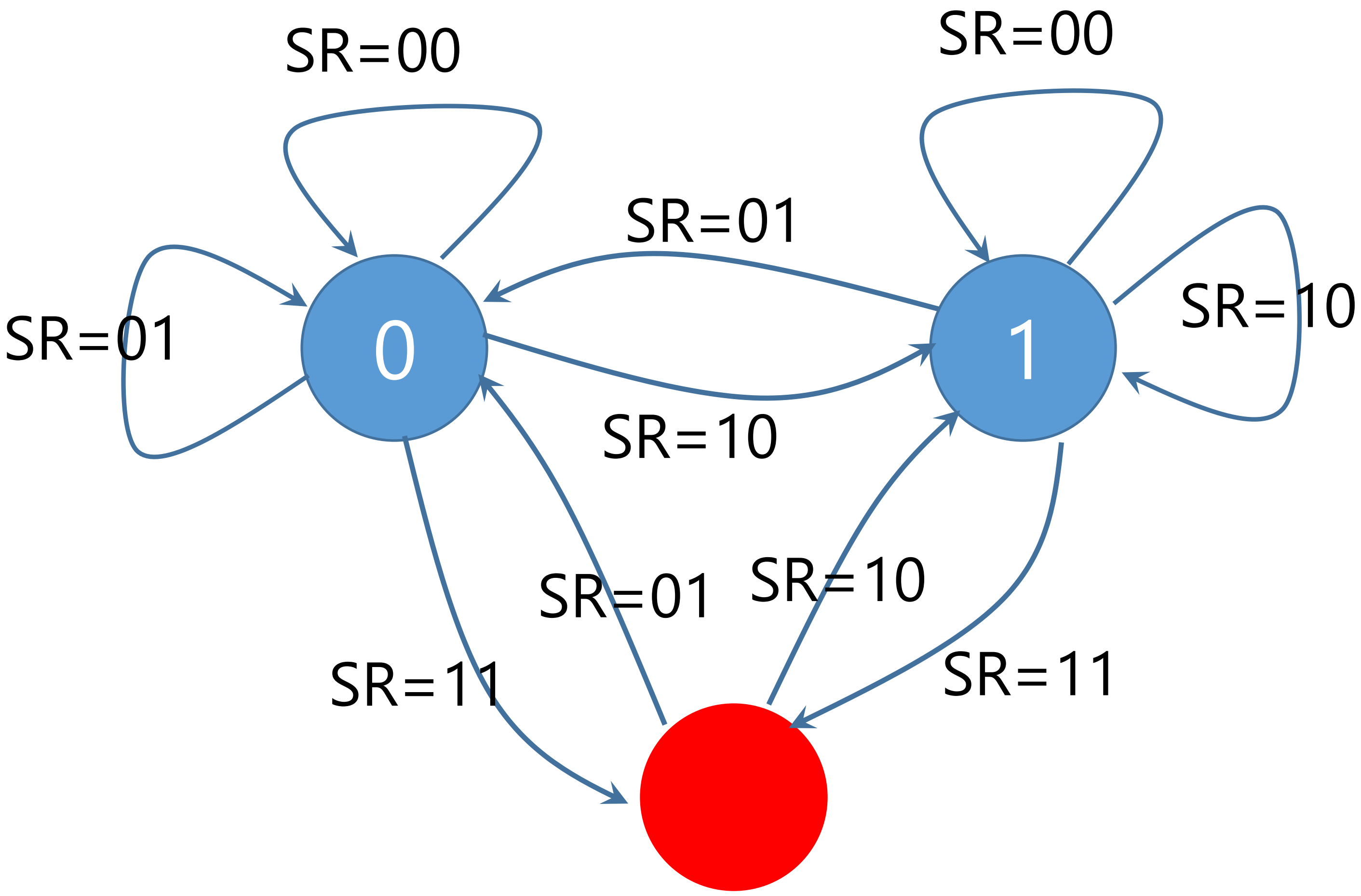
Life Cycle

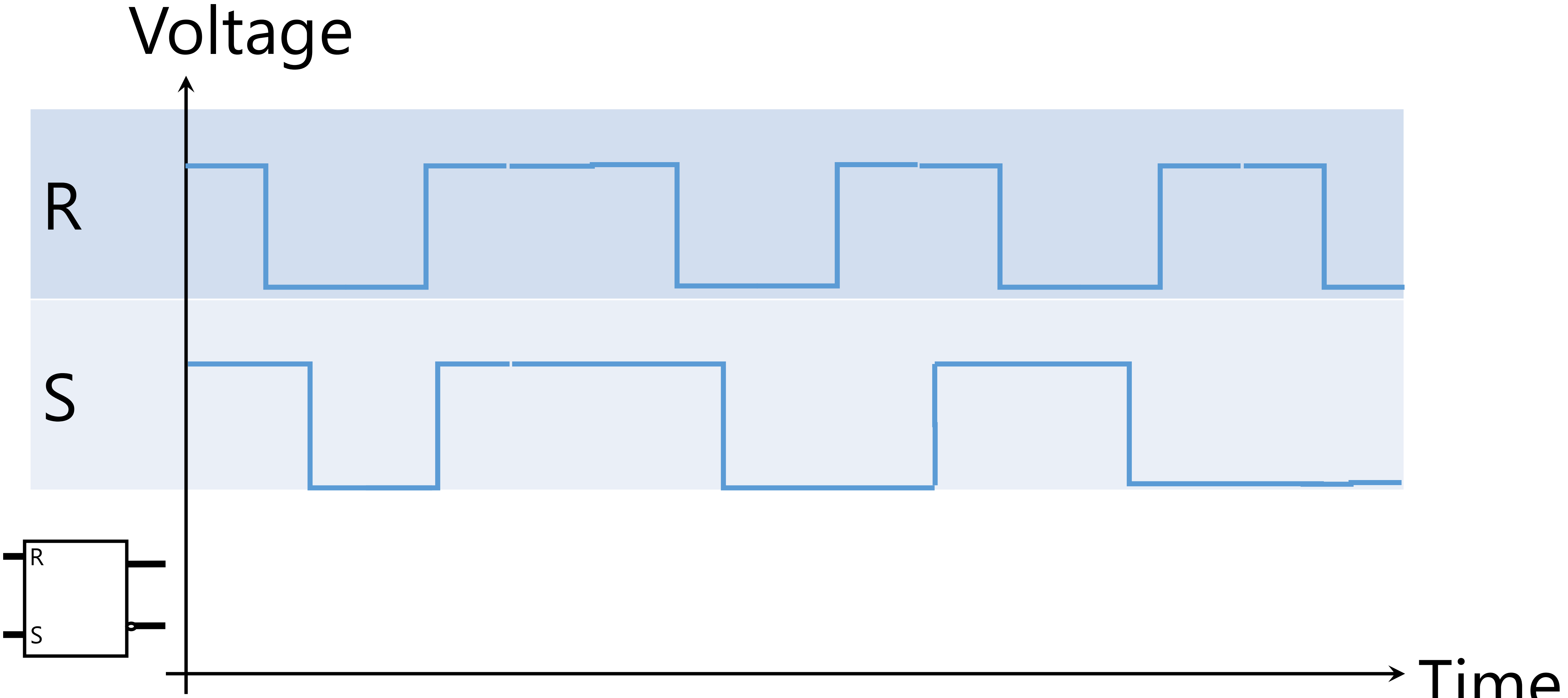
Block Diagram

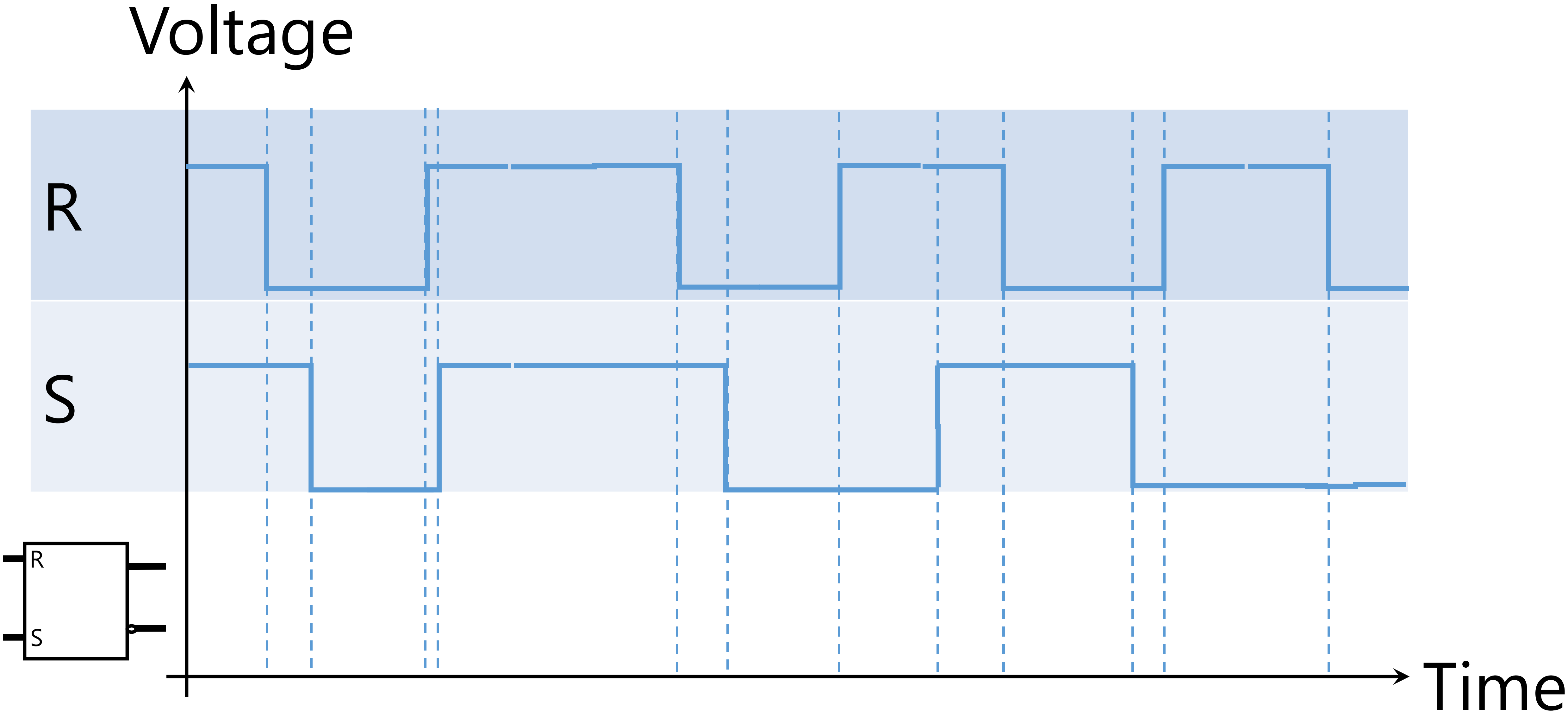


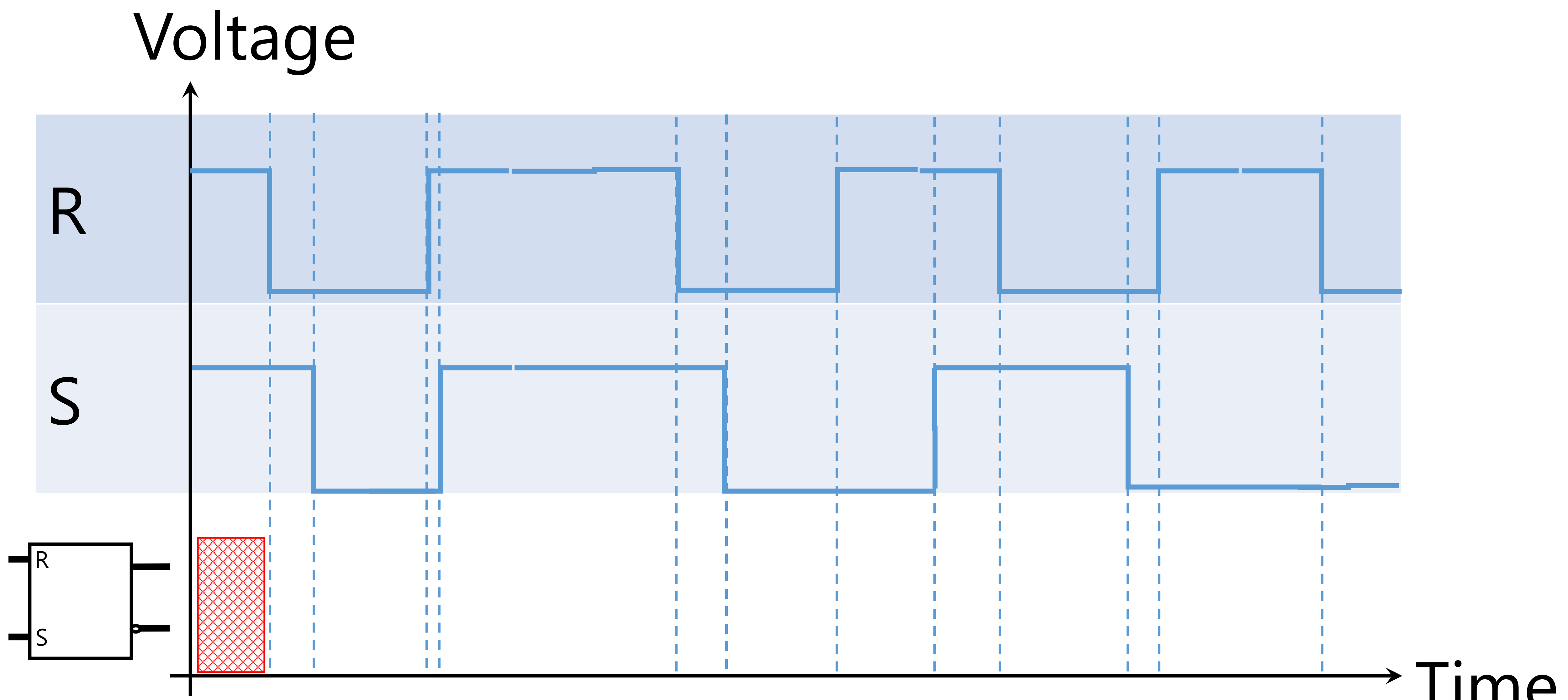
Characteristic Table

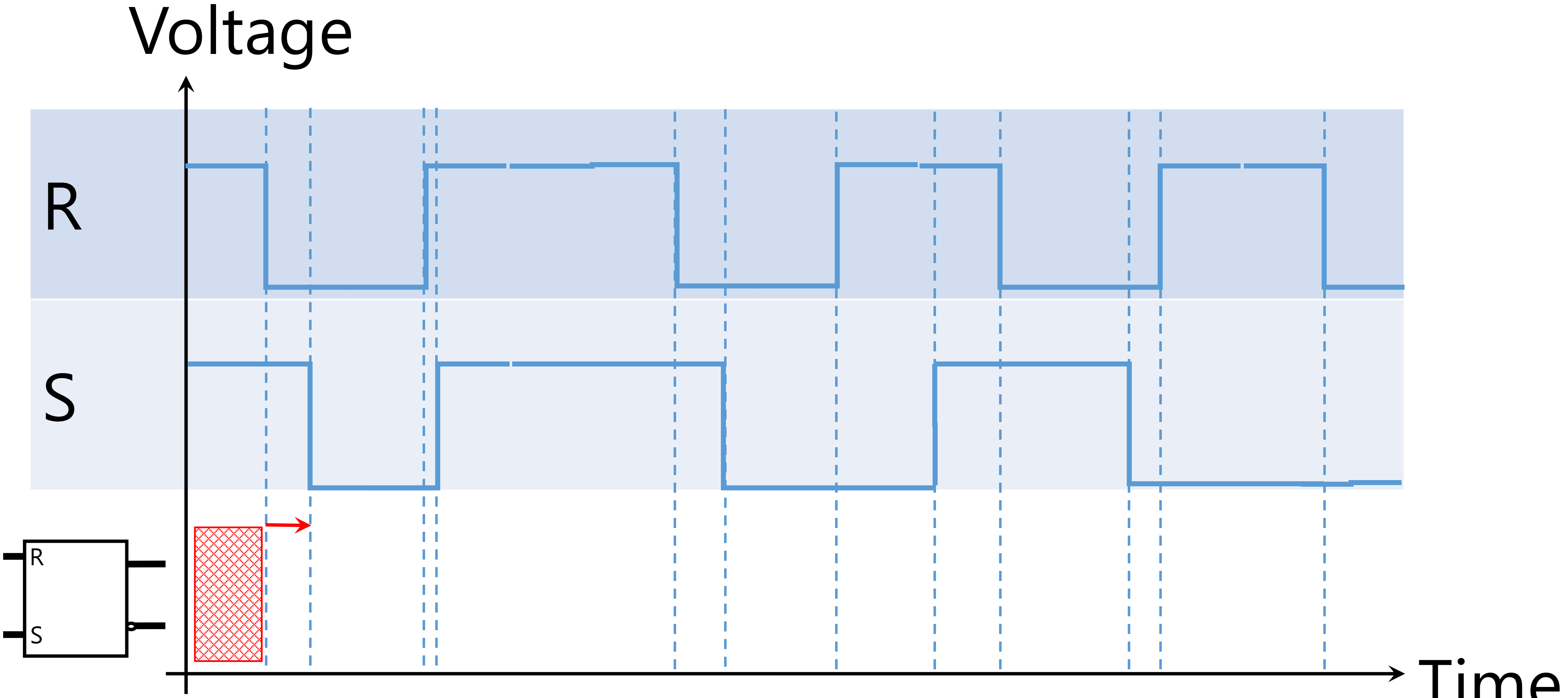
S	R	Q
0	0	Q_t
0	1	0
1	0	1
1	1	X



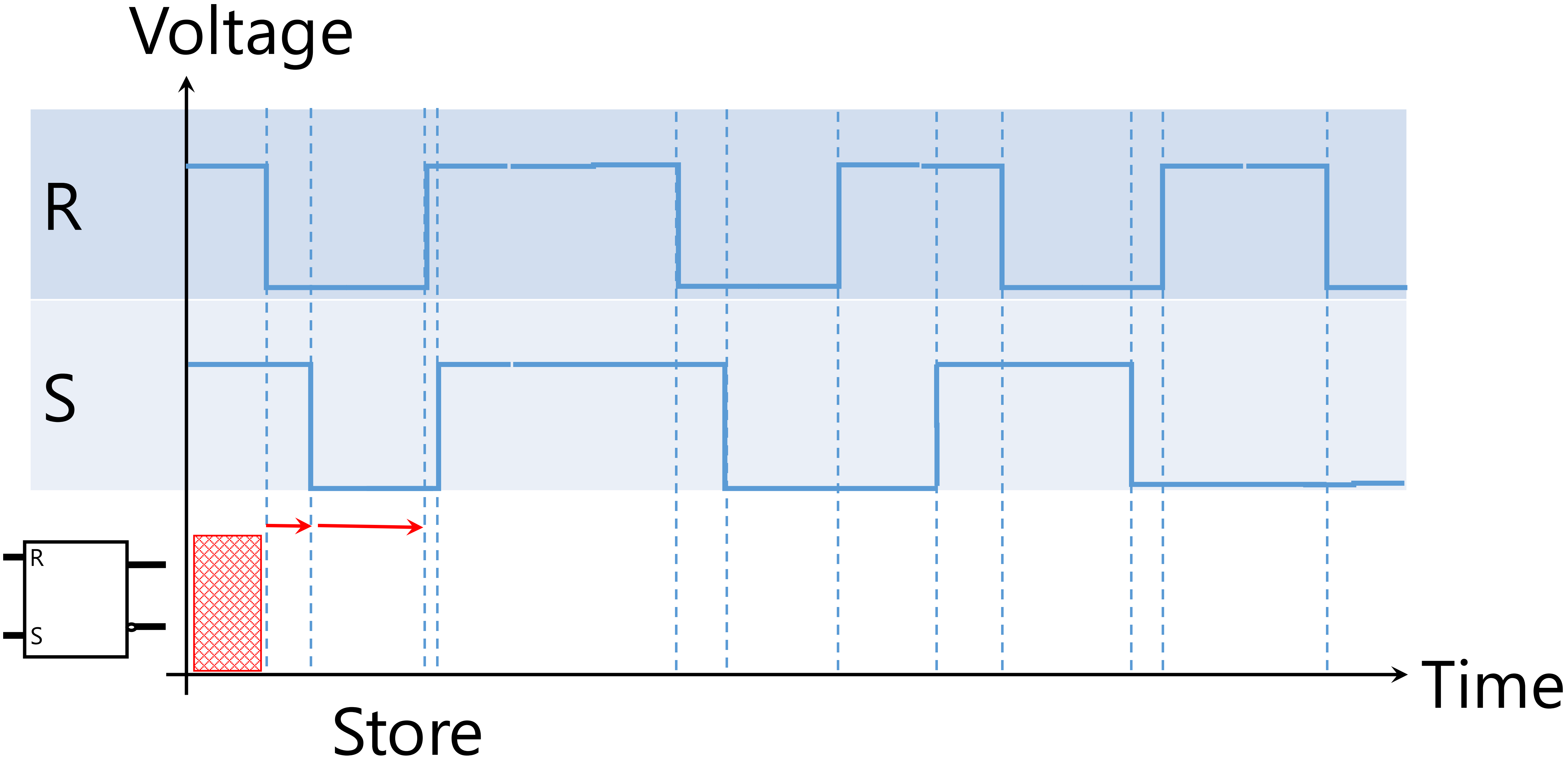

Voltage
R
S
R
S
Time

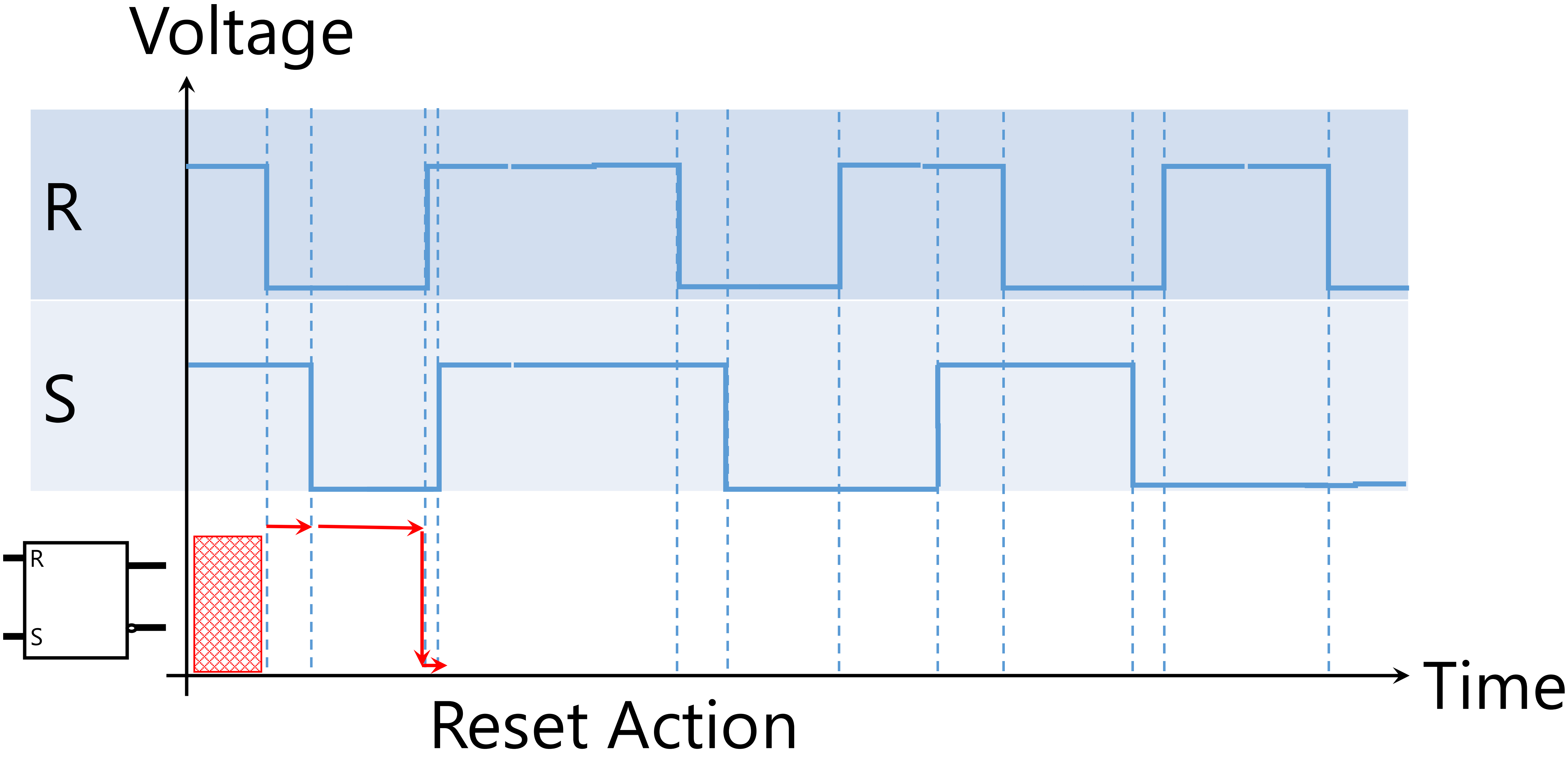

Voltage
R
S
R
S
Time


Voltage
R
S
R
S
Time

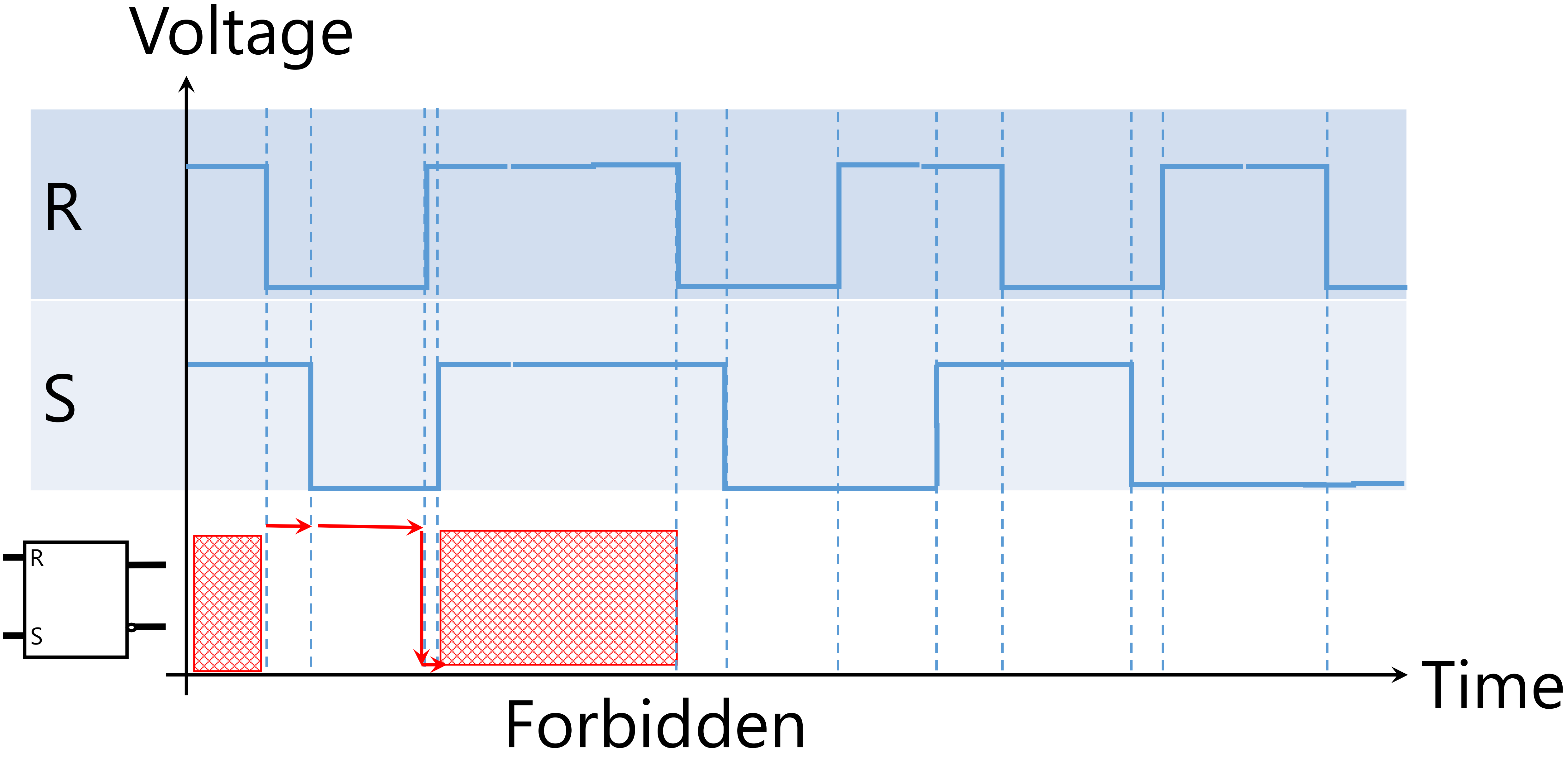


Set Action

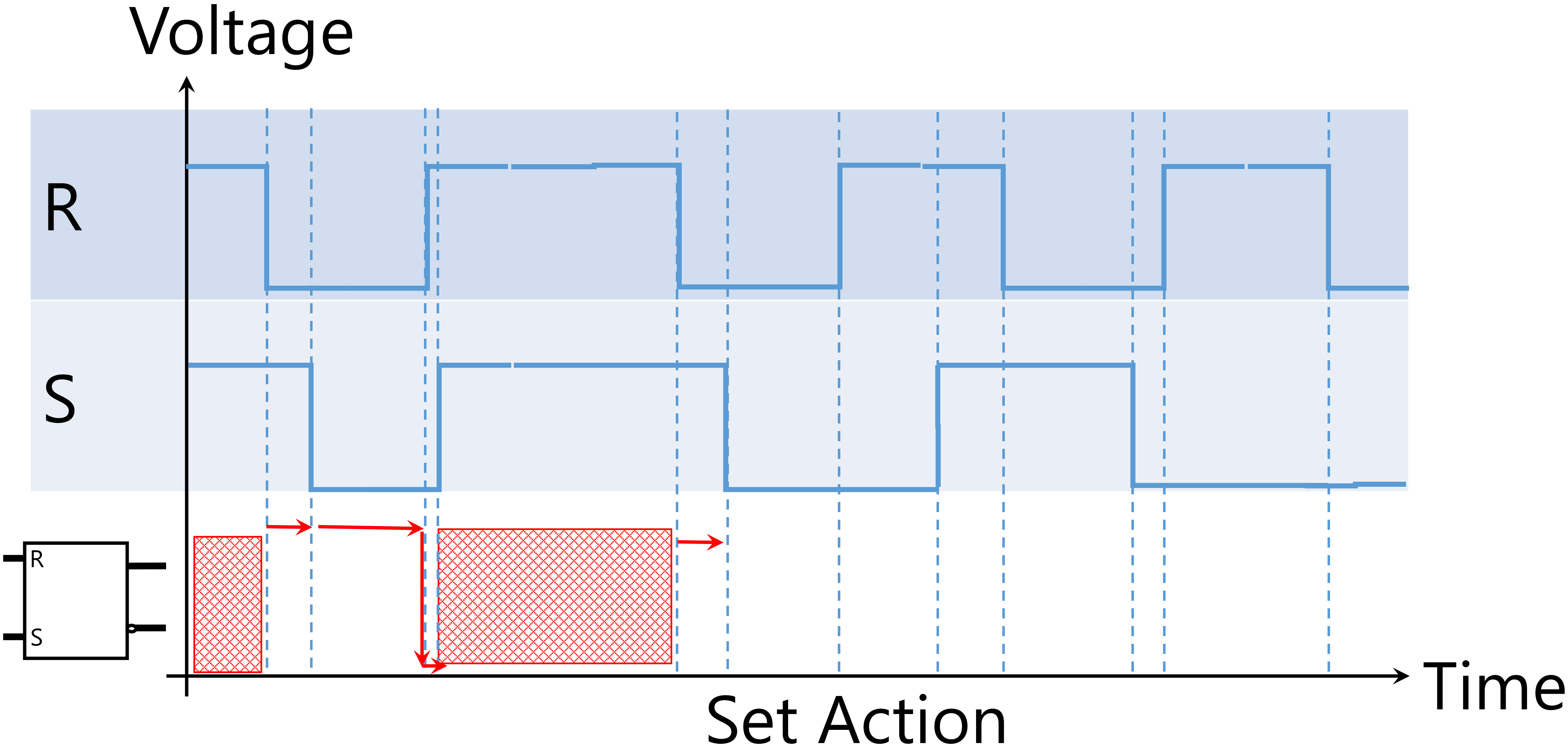

Voltage
R
S
R
S
Time
Store



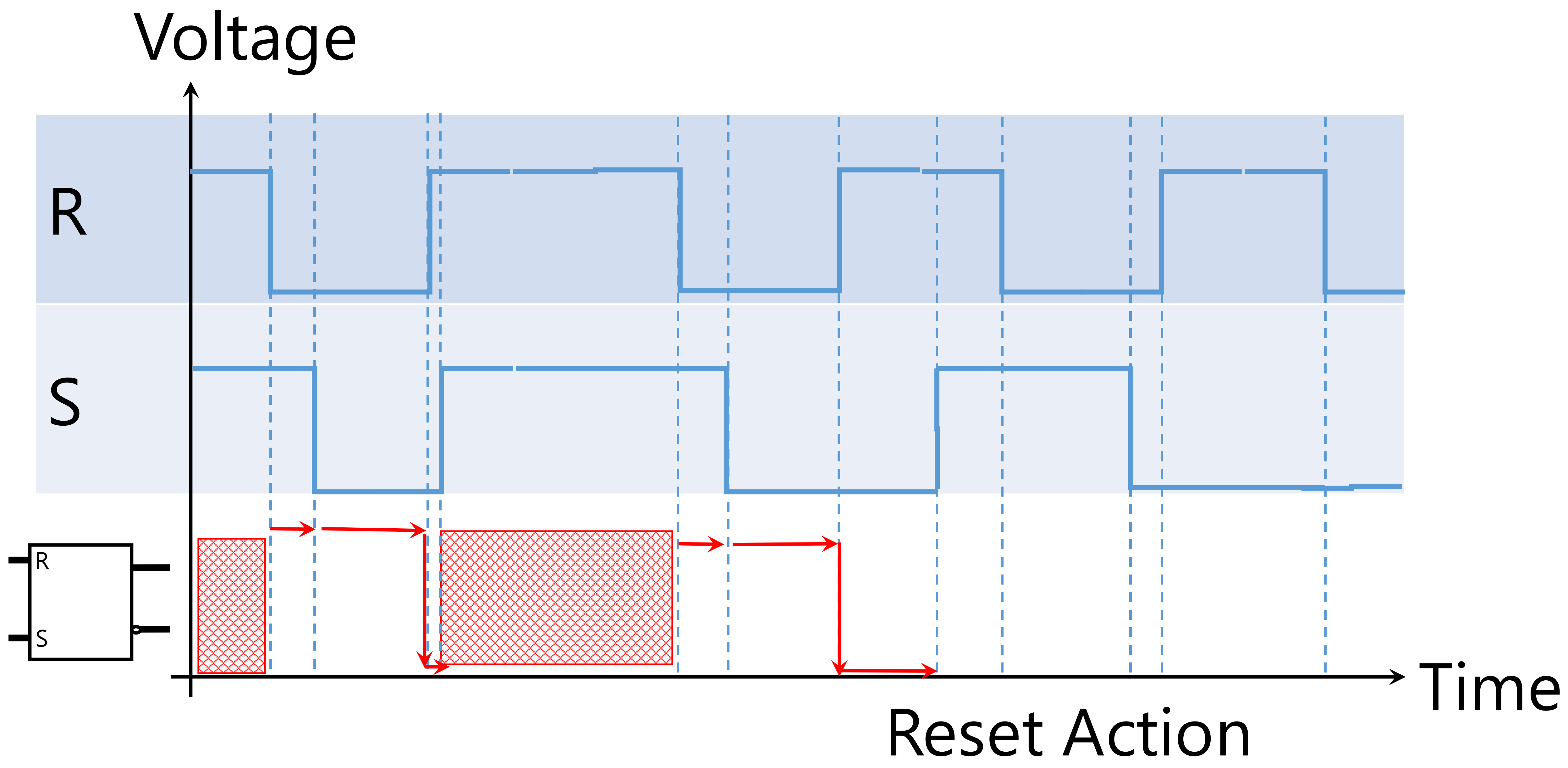
Voltage
R
S
R
S
Time
Reset Action



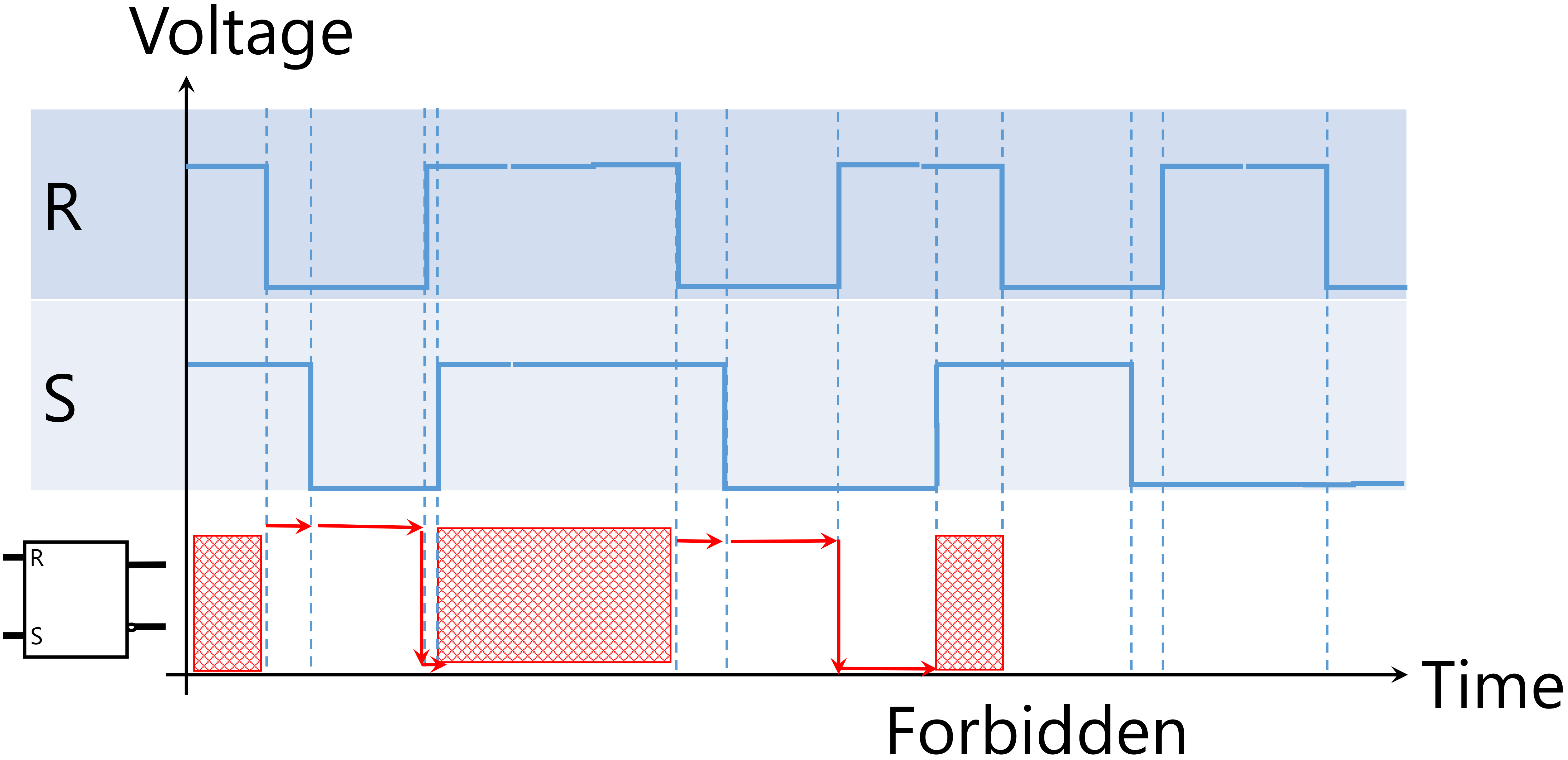
Voltage
R
S
R
S
Time
Forbidden

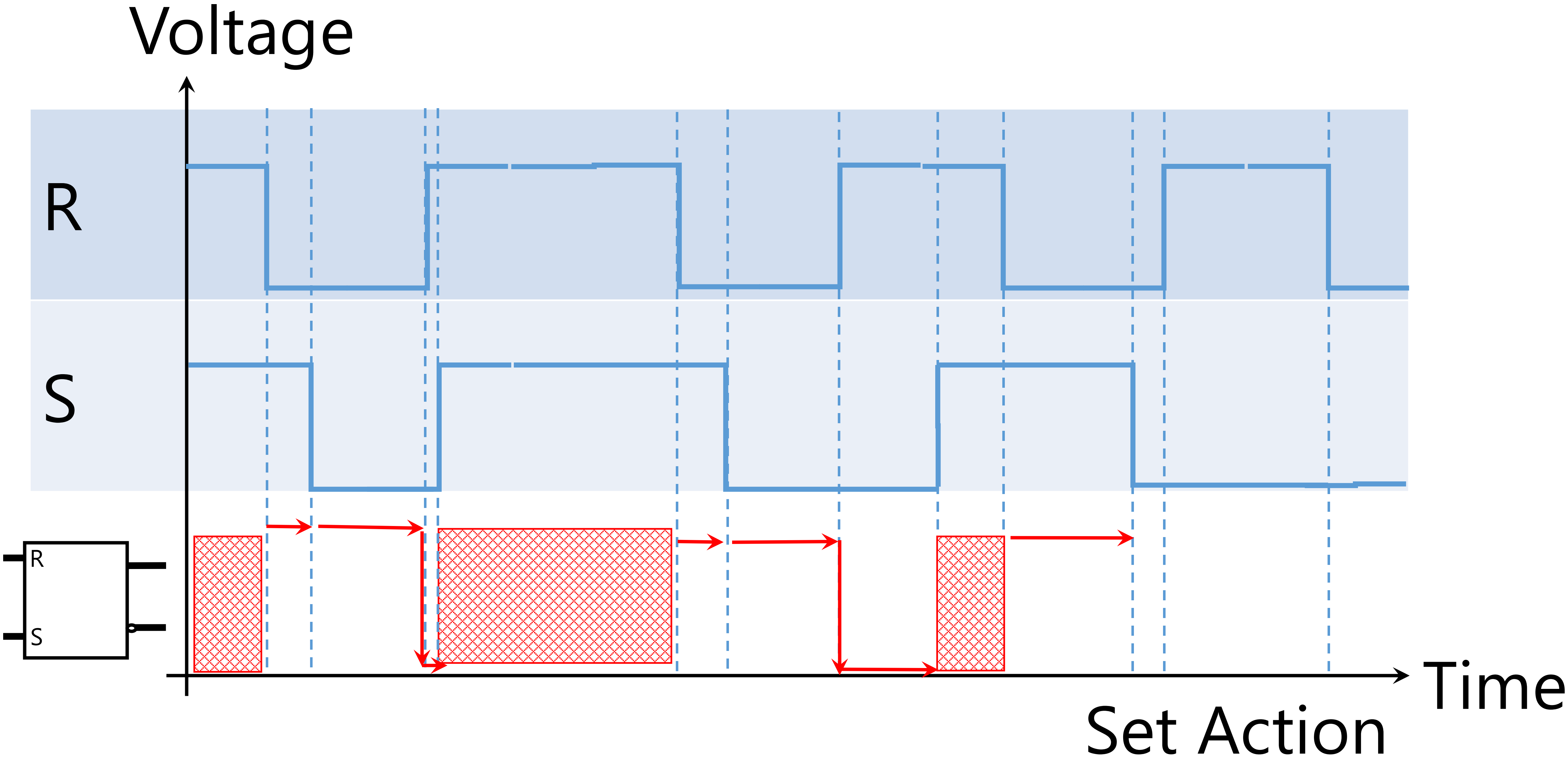

Voltage
R
S
R
S
Time
Set Action

Voltage
R
S
R
S
Time
Store Action

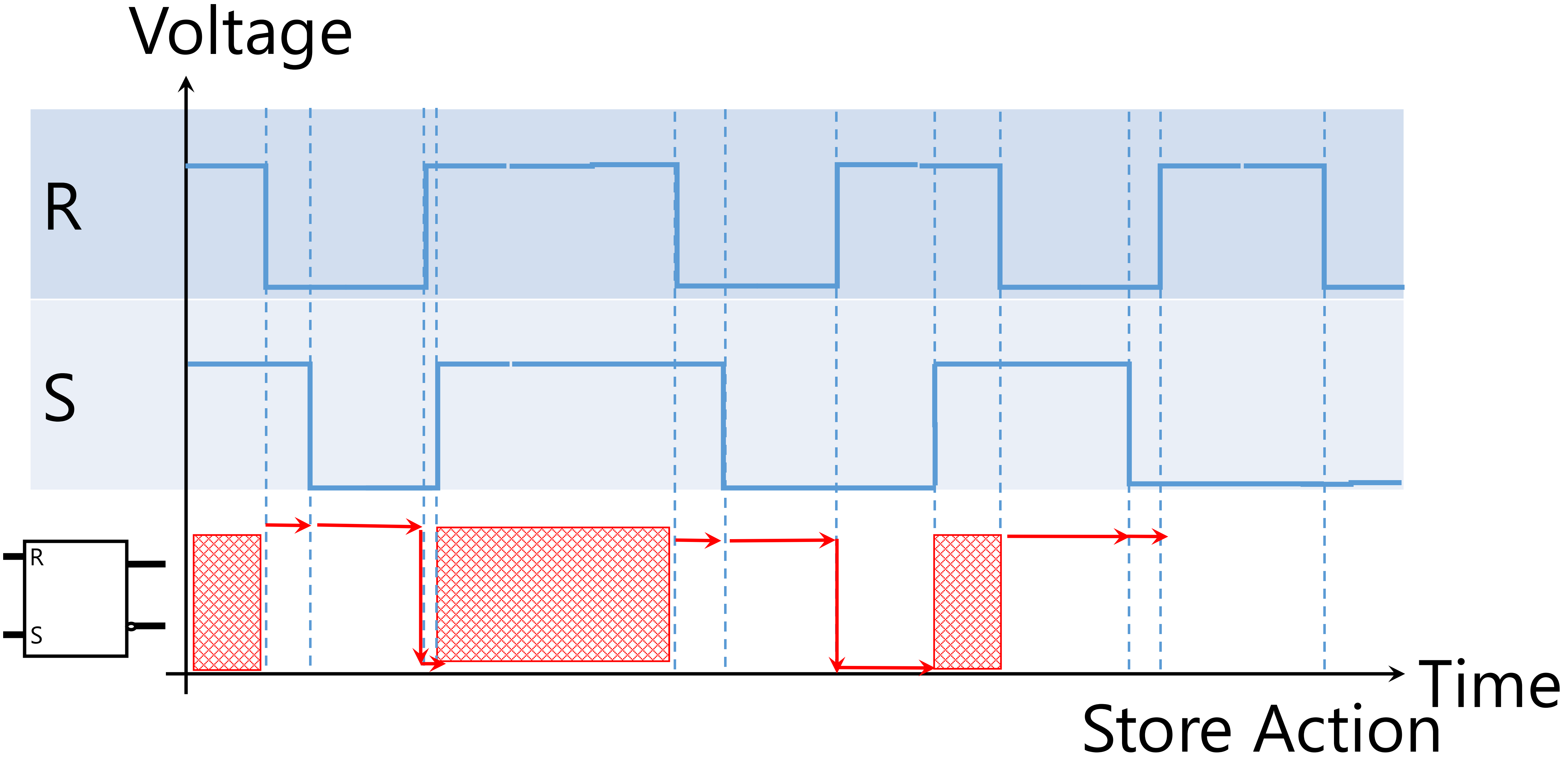


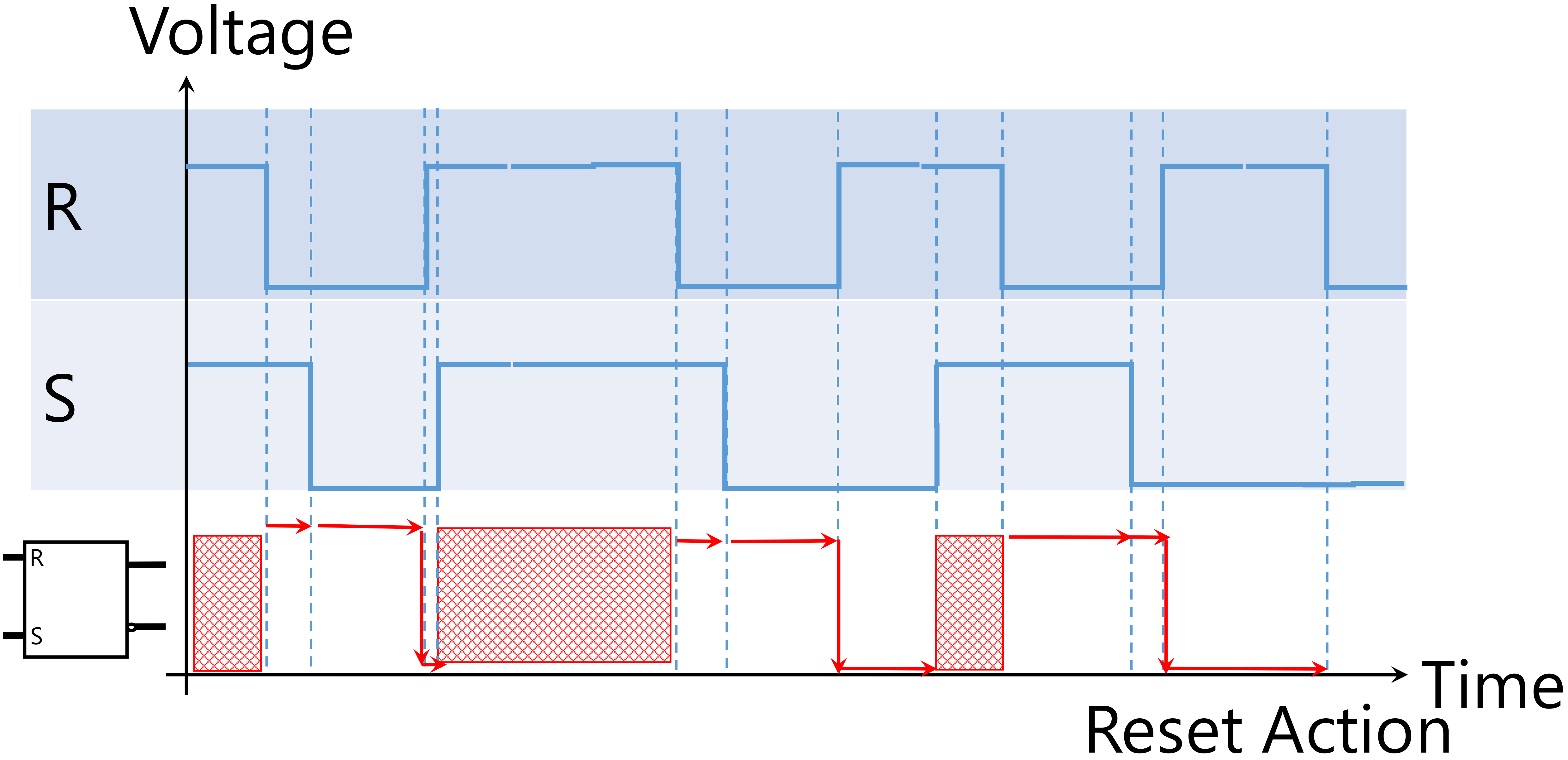
Voltage
R
S
R
S
Time
Reset Action


Voltage
R
S
R
S
Time
Forbidden

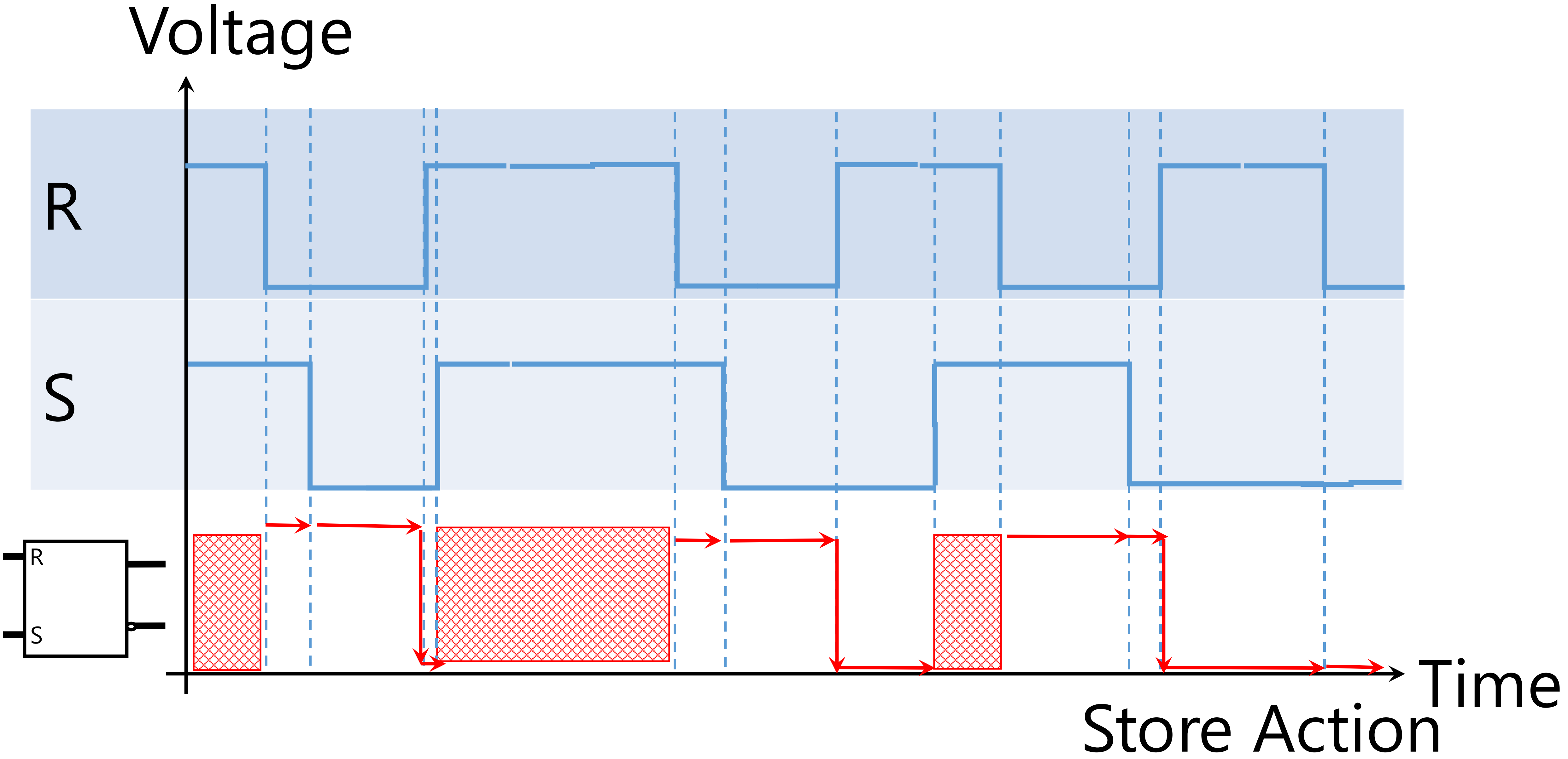


Voltage
R
S
R
S
Time
Set Action


Voltage
R
S
R
S
Time
Store Action



Voltage
R
S
R
S
Time
Reset Action

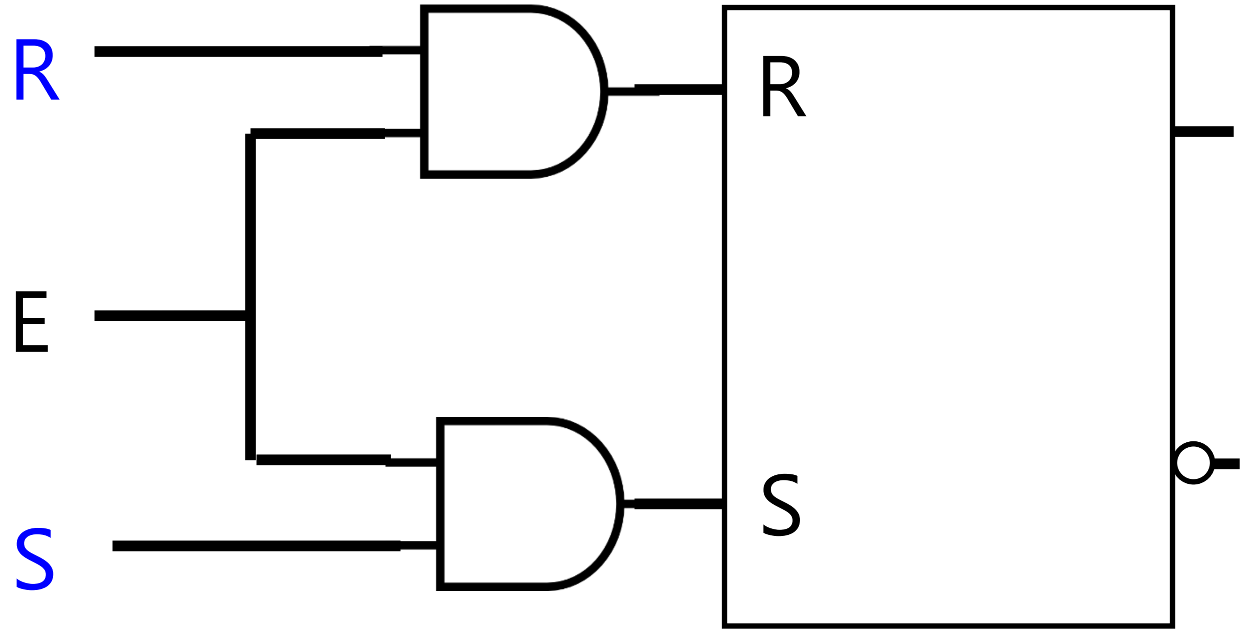

Voltage
R
S
R
S
Time
Store Action

S and R control how the state changes.

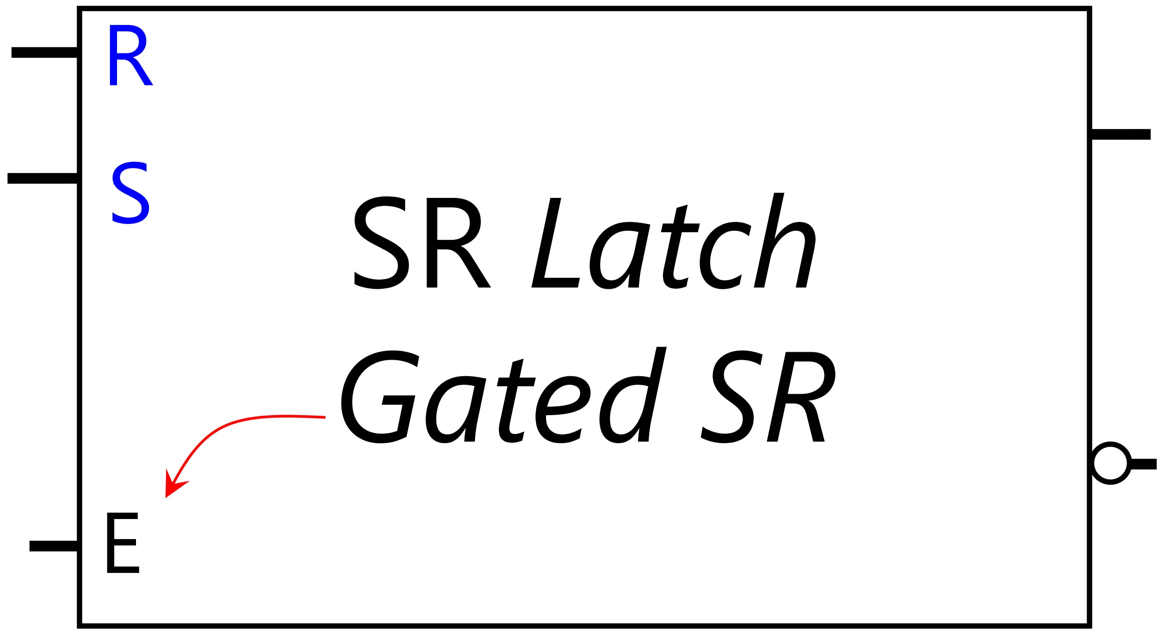
Put a gate/latch on when change applies
SR w/ enable input

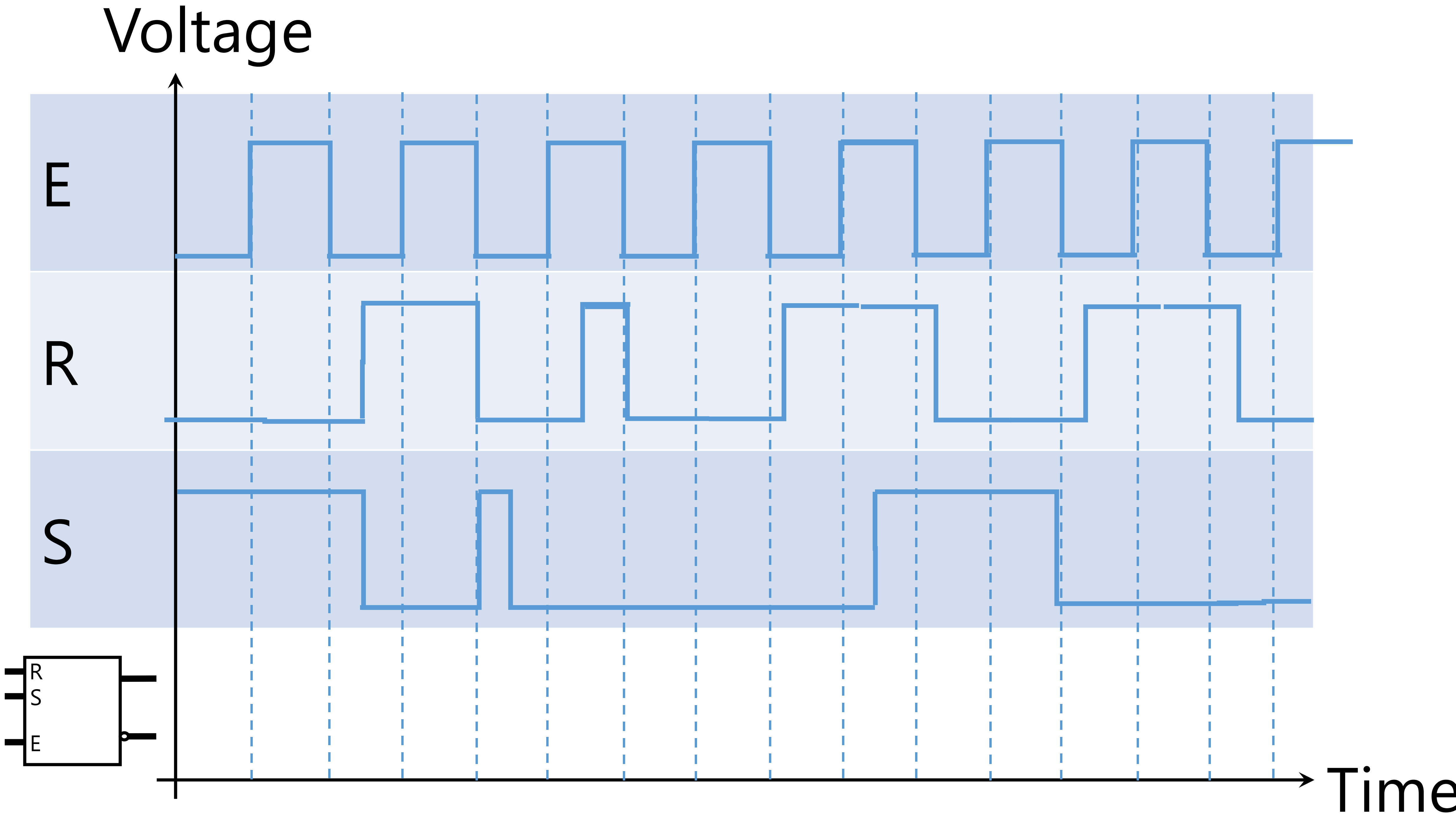
SR *Latch*

Gated SR

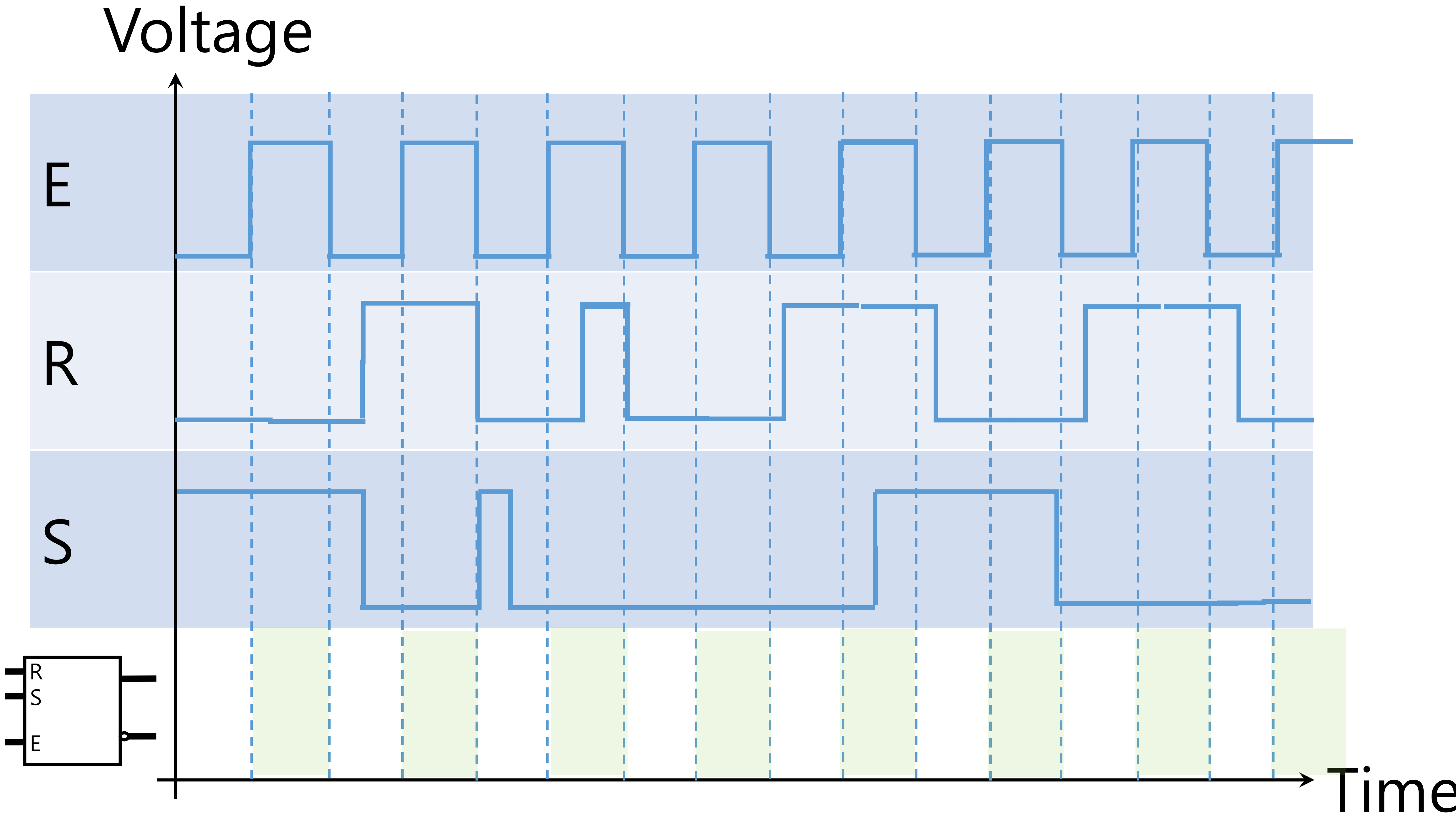


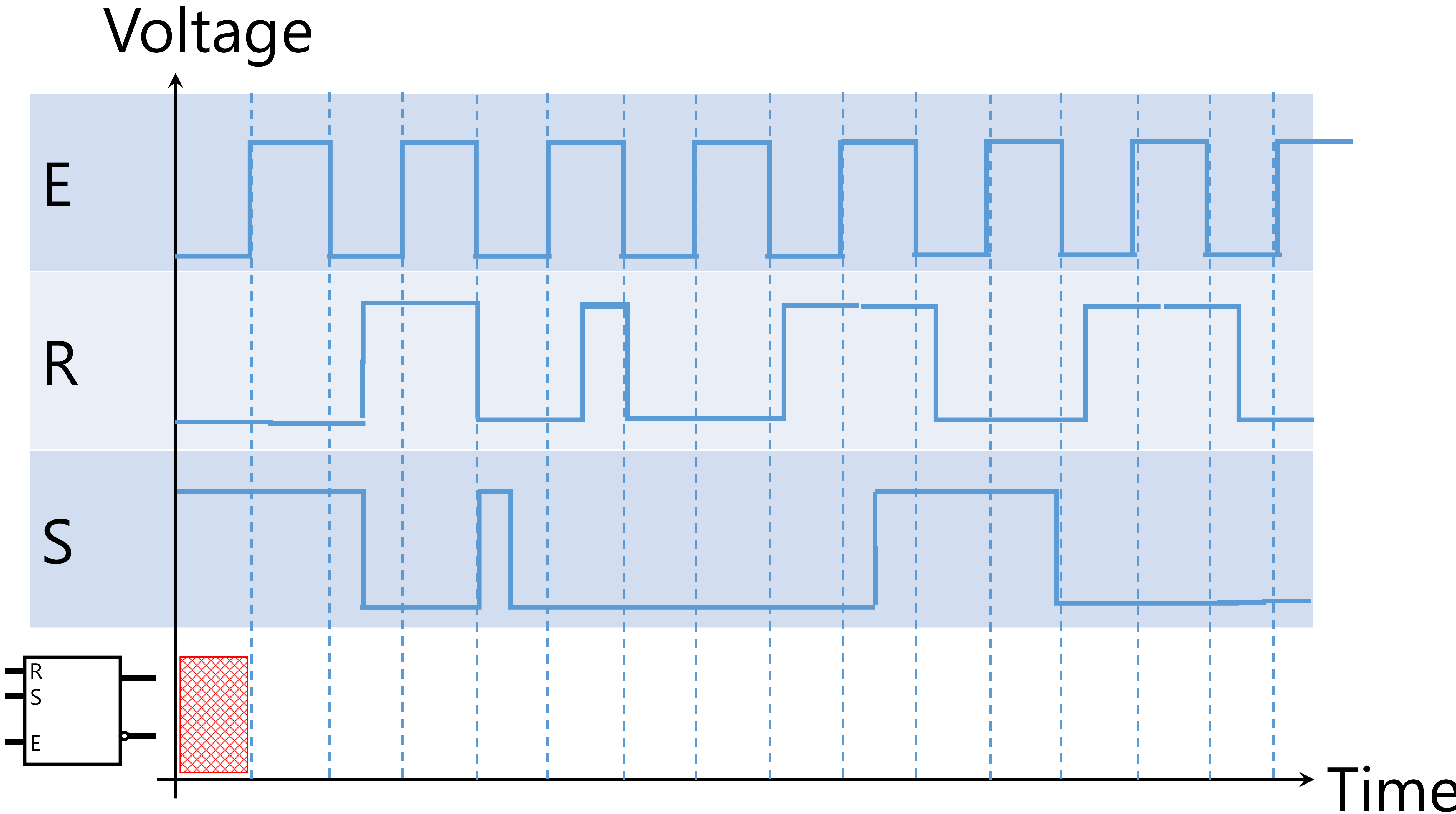
E	S	R	S	R	Q	Q′
0	x	x	0	0	Q_t	Q'_t
1	0	0	0	0	Q_t	Q'_t
1	0	1	0	1	0	1
1	1	0	1	0	1	0
1	1	1	1	1	x	x


R
S
SR Latch
Gated SR
E

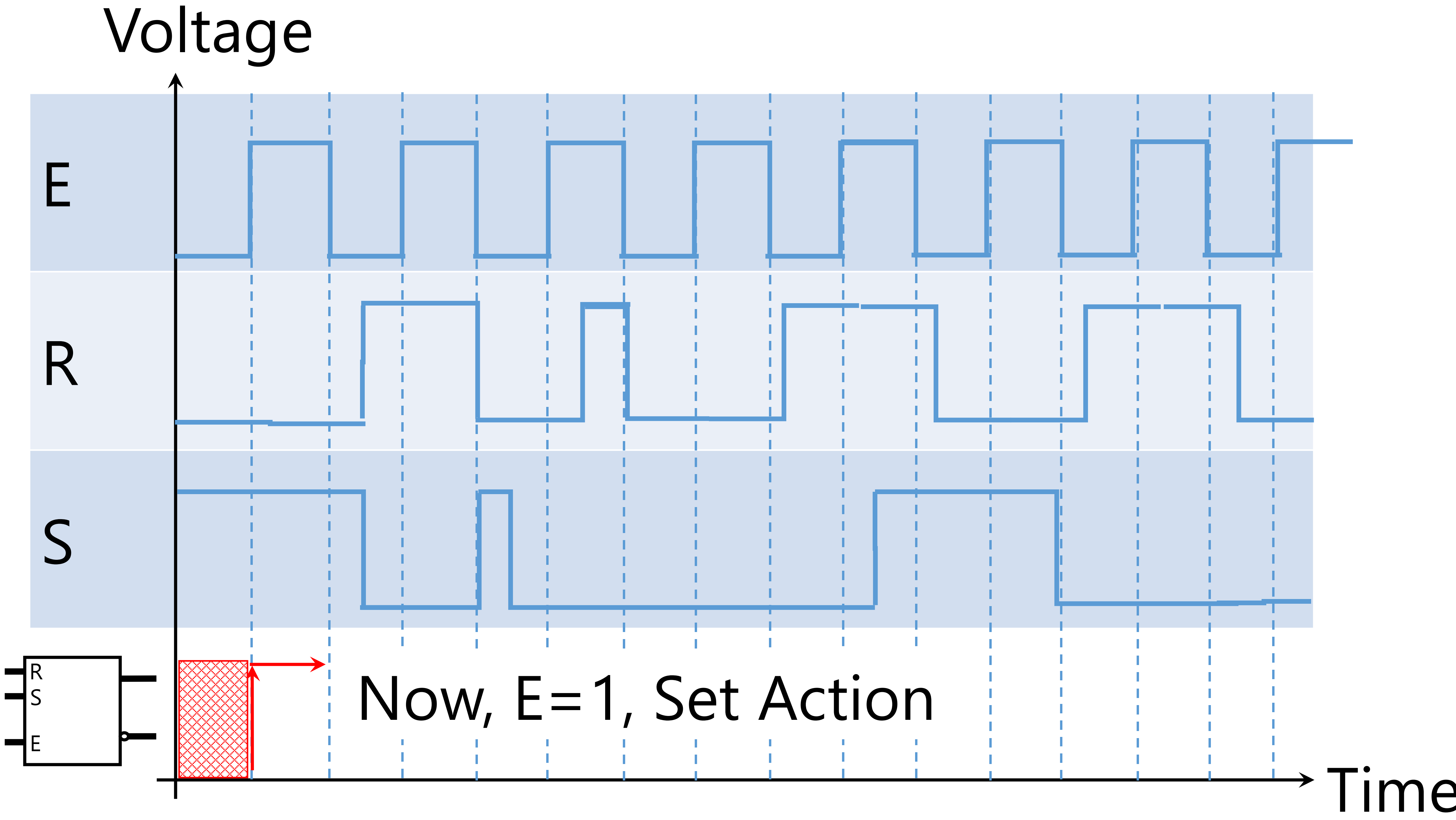


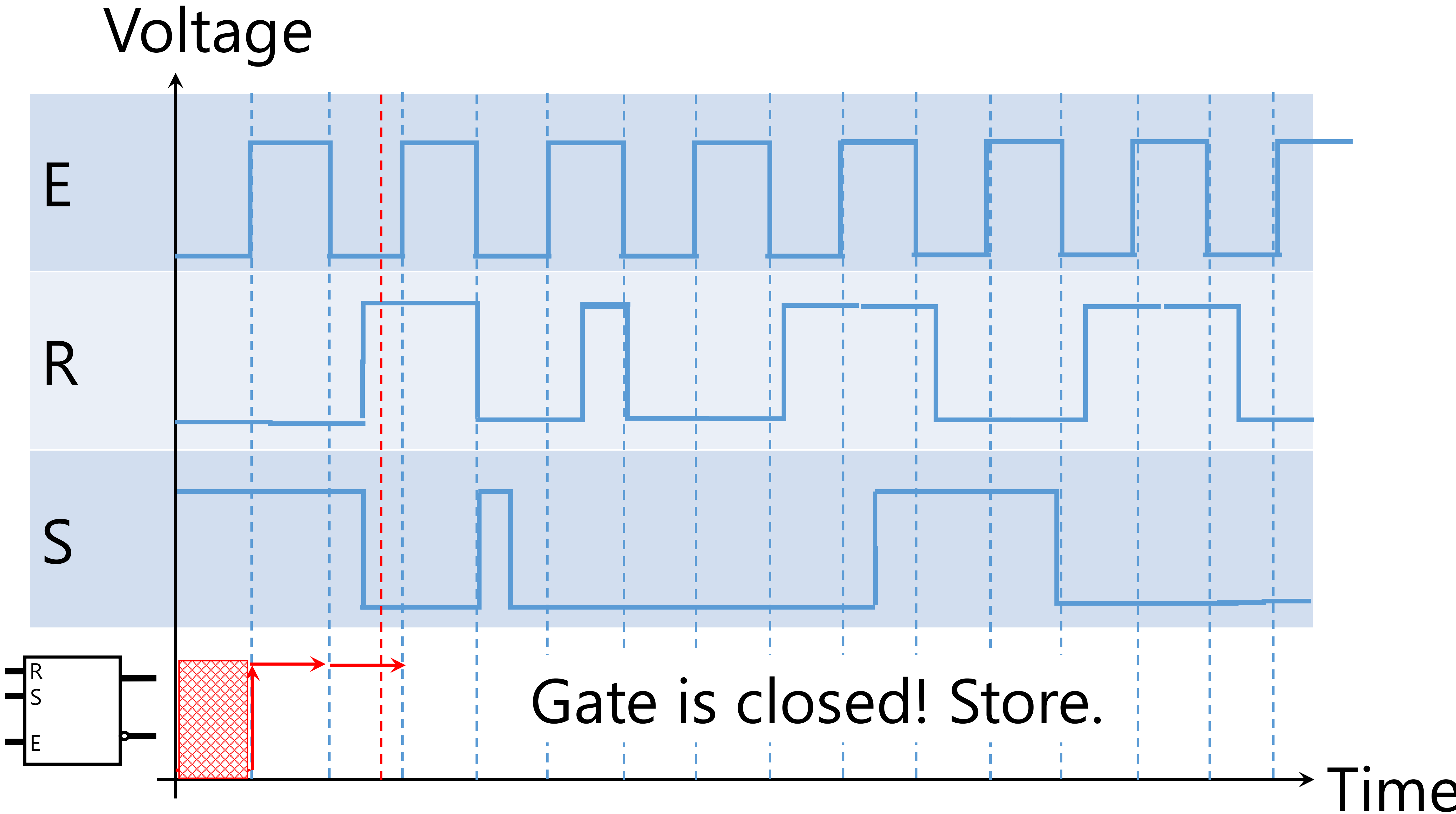
Voltage
E
R
S
R
S
E
Time


Voltage
E
R
S
R
S
E
Time

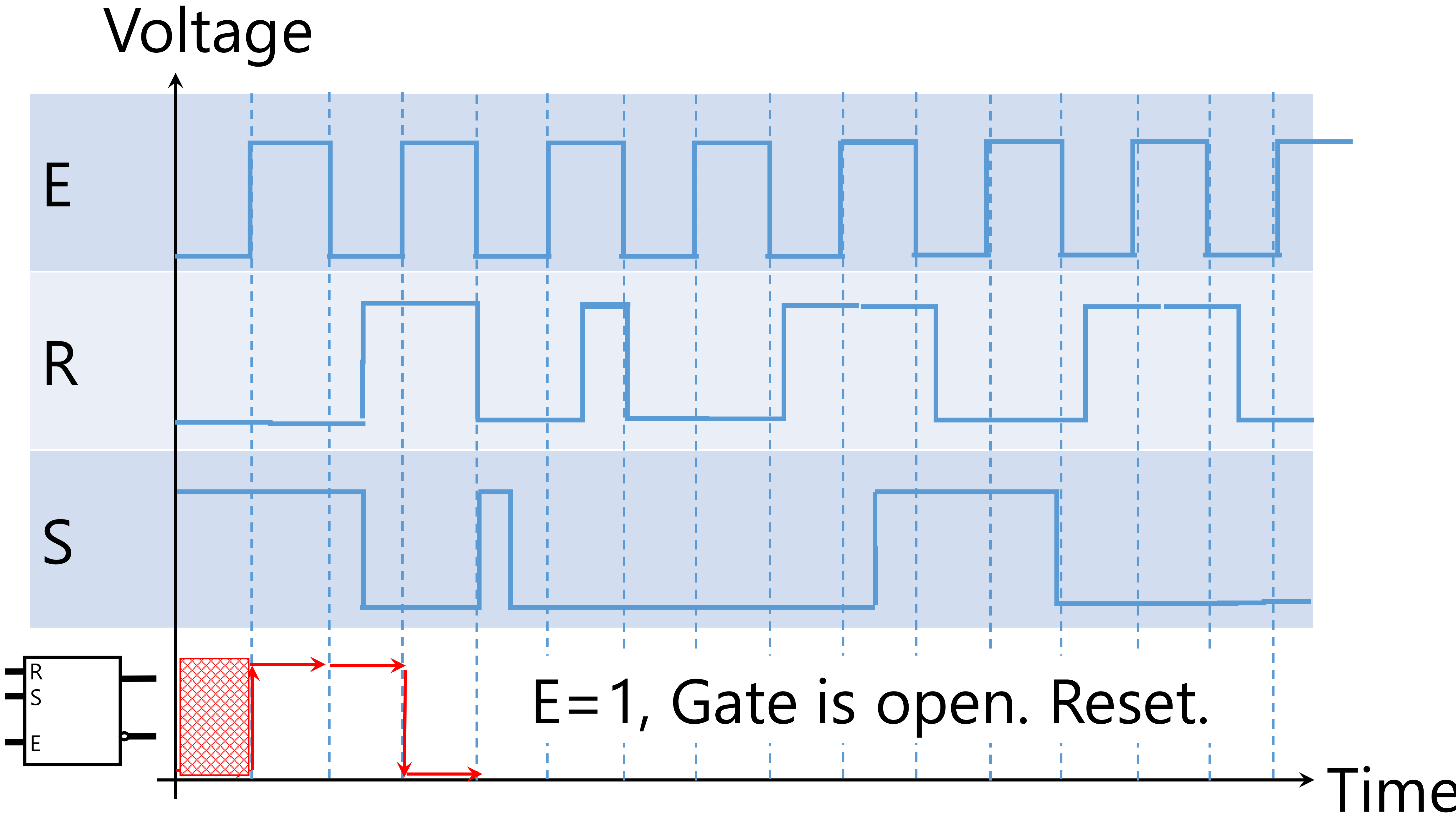


Voltage
E
R
S
R
S
E
Time

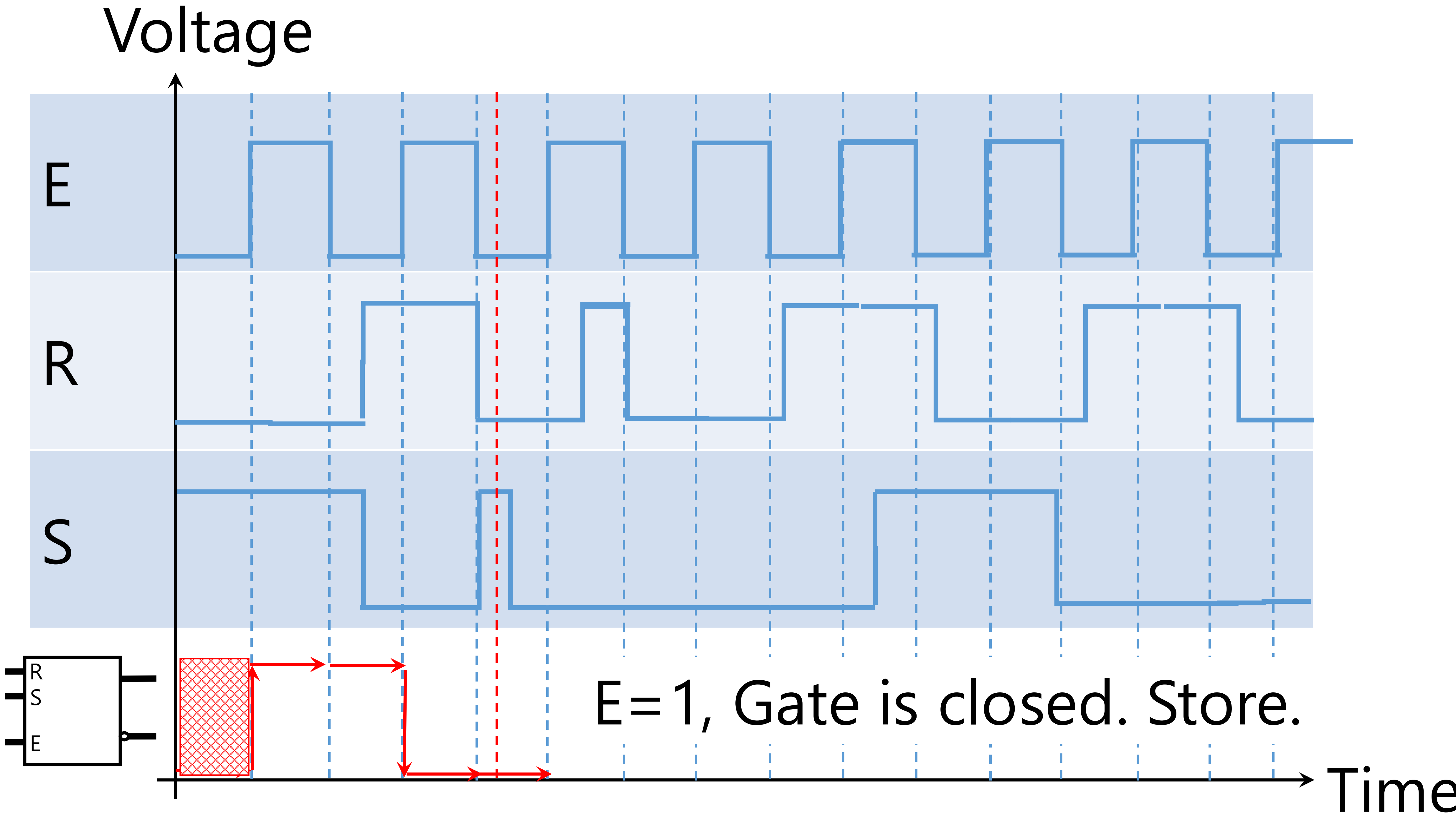

Voltage
E
R
S
R
S
E
Now, E=1, Set Action
Time



Voltage
E
R
S
R
S
E
Gate is closed! Store.
Time

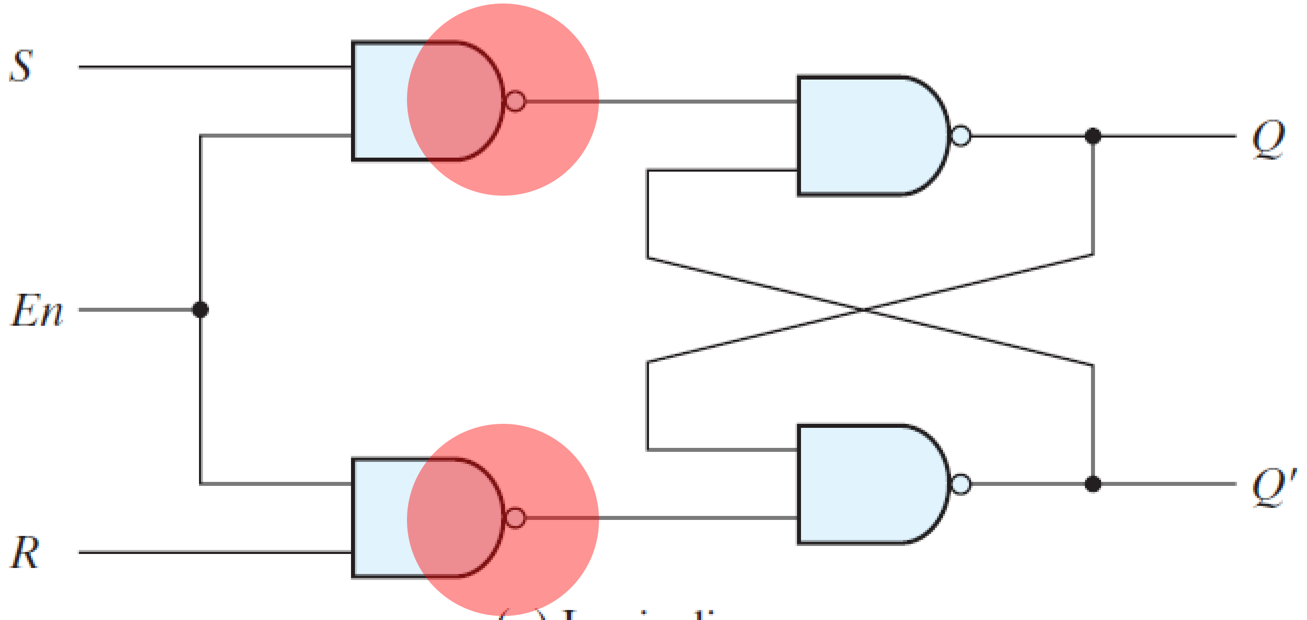


Voltage
E
R
S
R
S
E
E=1, Gate is open. Reset.
Time


Voltage
E
R
S
R
S
E
E=1, Gate is closed. Store.
Time

SR *Latch*

NAND

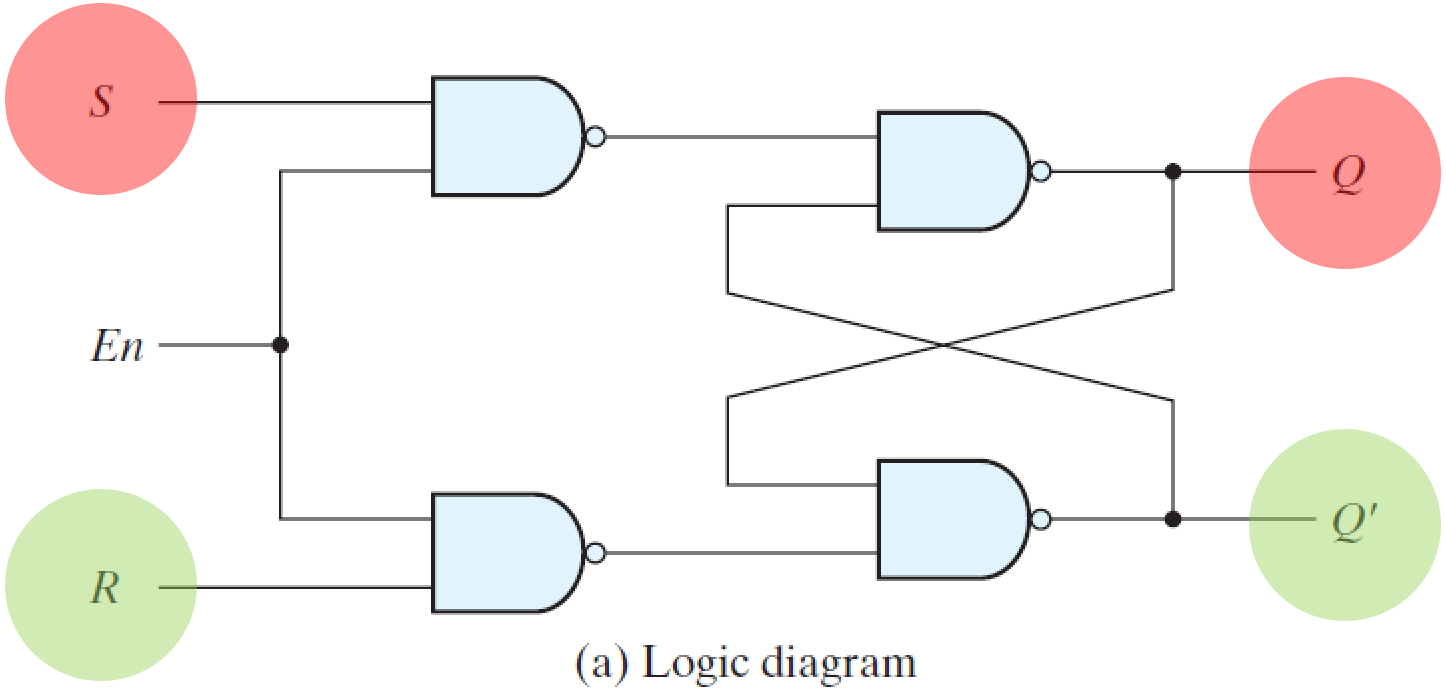


(a) Logic diagram

En	*S*	*R*	Next state of *Q*
0	X	X	No change
1	0	0	No change
1	0	1	$Q = 0$; reset state
1	1	0	$Q = 1$; set state
1	1	1	Indeterminate

(b) Function table

FIGURE 5.5
SR latch with control input



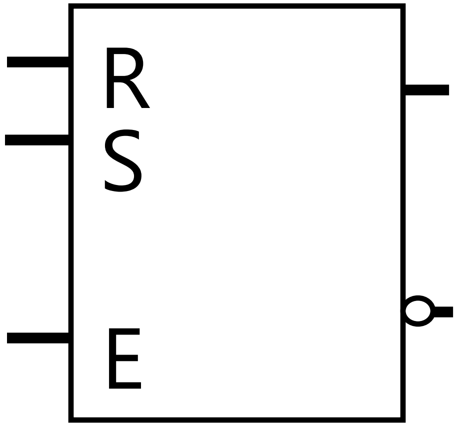
(a) Logic diagram

En	*S*	*R*	Next state of Q
0	X	X	No change
1	0	0	No change
1	0	1	$Q = 0$; reset state
1	1	0	$Q = 1$; set state
1	1	1	Indeterminate

(b) Function table

FIGURE 5.5
SR latch with control input

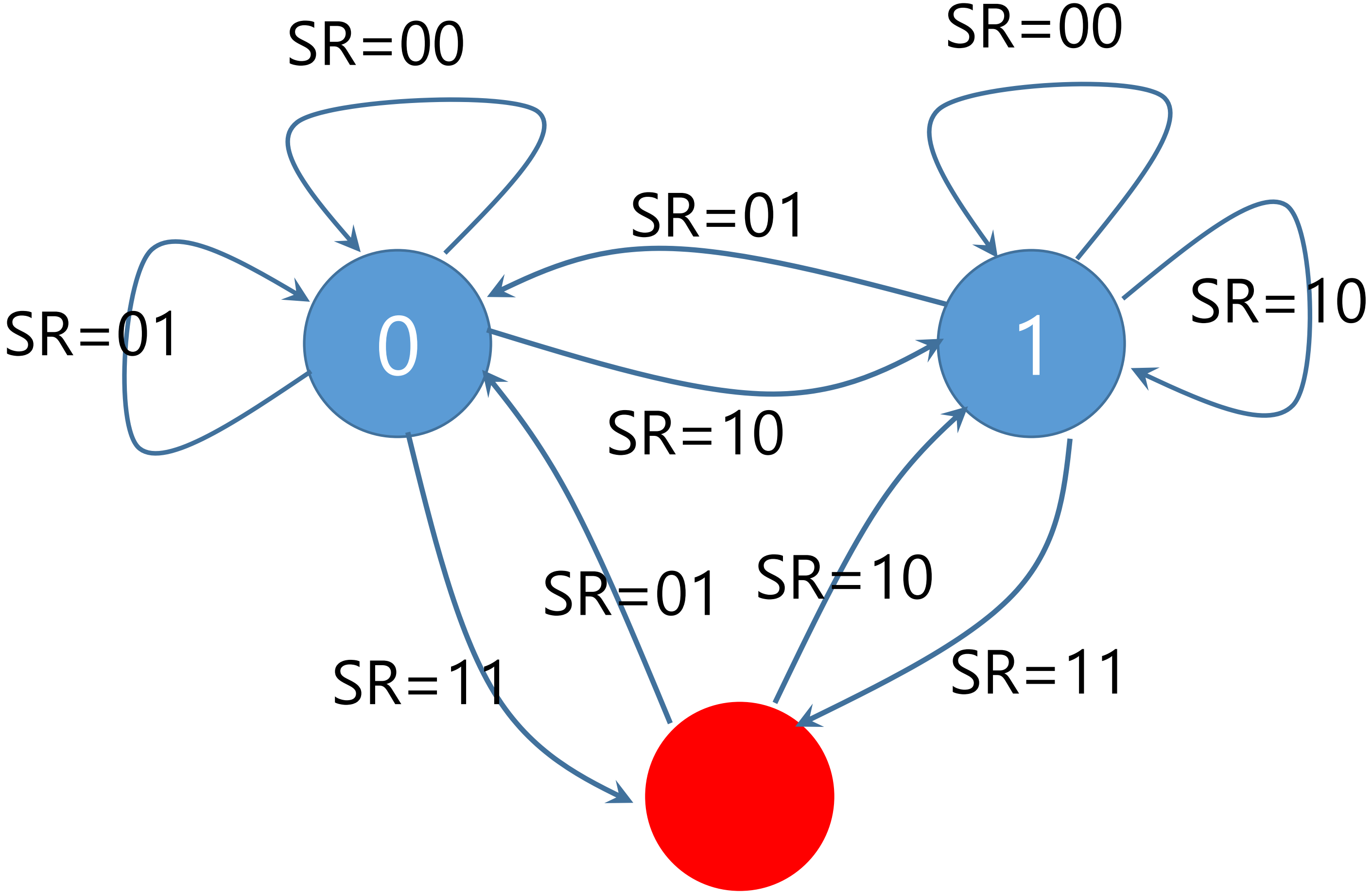
Block Diagram



Characteristic Table

S	R	Q
0	0	Q_t
0	1	0
1	0	1
1	1	X

State Transition Diagram



Other *Latches*